non-return valves without positive means of closing, provided that the inboard valve is situated above the deepest subdivision dra ught and is always accessible for examination under service conditions. Where a valve with positive means of closing is fitted, the operating position above the bulkhead deck shall always be readily accessible and means shall be provided for indicating whether the valve is open or closed.

8.2.2 The requirements of th e International Convention on Load Lines in force shall apply to discharges led through the shell pla ting from spaces above the bulkhead deck of passenger ships and the freeboard deck of cargo ships.

8.3 Machinery space, main and auxiliary sea inlets and discharges in connection with the operation of machinery shall be fitted with readily accessible valves between the pipes

8.3 Machinery space, main and auxiliary sea inlets and discharges in connection with the operation of machinery shall be fitted with readily accessible valves between the pipes and the shell plating or betw een the pipes and fabricated boxes attached to the shell plating. In manned machinery spaces the valves may be controlled locally and shall be provided with indicators showing whether they are open or closed.

8.4 Moving parts penetrating the shell plating below the deepest subdivision draught shall be fitted with a watert ight sealing arrangement acceptable to the Administration. The inboard gland shall be located within a watertight space of such volume that, if flooded, the bulkhead deck will not be submer ged. The Administration may require that if such compartment is flooded, essential or emergency power and lighting, internal communication, signals or other emergency devi ces must remain available in other parts of the ship.

if such compartment is flooded, essential or emergency power and lighting, internal communication, signals or other emergency devi ces must remain available in other parts of the ship.

8.5 All shell fittings and valves required by this regulation sh all be of steel, bronze or other approved ductile material. Valves of or dinary cast iron or similar material are not acceptable. All pipes to which this regulation re fers shall be of steel or other equivalent material to the satisfaction of the Administration.

9 Gangway, cargo and fuelling ports fitted below the bulkhead deck of passenger ships and the freeboard deck of cargo ships shall be watertight and in no case be so fitted as to have their lowest point below the deepest subdivision draught.

10.1 The inboard opening of each ash-chute, r ubbish-chute, etc., shall be fitted with an efficient cover.

10.2 If the inboard opening is situated below the bulkhead deck of passenger ships and

10.1 The inboard opening of each ash-chute, r ubbish-chute, etc., shall be fitted with an efficient cover.

10.2 If the inboard opening is situated below the bulkhead deck of passenger ships and the freeboard deck of cargo ships, the cover shall be watertight and, in addition, an automatic non-return valve shall be fitted in the chute in an easily accessible position above the deepest subdivision draught.

Regulation 15-1
External openings in cargo ships

1 All external openings leading to compar tments assumed intact in the damage analysis, which are below the final damage waterline, are required to be watertight.

2 External openings required to be waterti ght in accordance with paragraph 1 shall, except for cargo hatch covers, be fitted with indicators on the bridge.

3 Openings in the shell plati ng below the deck limiting the vertical extent of damage shall be fitted with a device that prevents unauthorized op ening if they are accessible during the voyage.

3 Openings in the shell plati ng below the deck limiting the vertical extent of damage shall be fitted with a device that prevents unauthorized op ening if they are accessible during the voyage.

4 Other closing appliances which are kept permanently closed at sea to ensure the watertight integrity of external openings sh all be provided with a notice affixed to each appliance to the effect that it is to be ke pt closed. Manholes fitted with closely bolted covers need not be so marked.

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Regulation 16
Construction and initial tests of watertight doors, sidescuttles, etc.

1 In all ships:

.1 the design, materials and construction of all watertight doors, sidescuttles, gangway and cargo ports, valves, pipes, ash-chut es and rubbish-chutes referred to in these regulations shall be to the satisfaction of the

gangway and cargo ports, valves, pipes, ash-chut es and rubbish-chutes referred to in these regulations shall be to the satisfaction of the Administration;

.2 such valves, doors and mechanisms shall be suitably marked to ensure that they may be properly used to provide maximum safety; and

.3 the frames of vertical watertight doors shall have no groove at the bottom in which dirt might lodge and prevent the door closing properly.

2 In passenger ships and cargo ships wate rtight doors shall be tested by water pressure to a head of water they might sustain in a final or intermediate stage of flooding. Where testing of individual doors is not carried out because of possible damage to insulation or outfitting items, testing of i ndividual doors may be replaced by a prototype pressure test of each type and size of door w ith a test pressure corresponding at least to the head required for the intended location. Th e prototype test shall be carried out before

pressure test of each type and size of door w ith a test pressure corresponding at least to the head required for the intended location. Th e prototype test shall be carried out before the door is fitted. The installation met hod and procedure for fitting the door on board shall correspond to that of the prototype te st. When fitted on board, each door shall be checked for proper seating between the bulkhead, the frame and the door.

Regulation 16-1
Construction and initial tests of watertight decks, trunks, etc.

1 Watertight decks, trunks, tunnels, duct keels and ventilators shall be of the same strength as watertight bulkheads at corres ponding levels. The means used for making them watertight, and the arrangements adopted for closing openings in them, shall be to the satisfaction of the Administ ration. Watertight ventilator s and trunks shall be carried at least up to the bulkhead deck in passenge r ships and up to the freeboard deck in cargo ships.

the satisfaction of the Administ ration. Watertight ventilator s and trunks shall be carried at least up to the bulkhead deck in passenge r ships and up to the freeboard deck in cargo ships.

2 Where a ventilation trunk passing thr ough a structure penetrates the bulkhead deck, the trunk shall be capable of withstandi ng the water pressure that may be present within the trunk, after havi ng taken into account the maximum heel angle allowable during intermediate stages of flooding, in accordance with regulation 7-2.

3 Where all or part of the penetration of the bulkhead deck is on the main ro-ro deck, the trunk shall be capable of withstandi ng impact pressure due to internal water motions (sloshing) of water trapped on the ro-ro deck.

4 After completion, a hose or flooding test sh all be applied to watertight decks and a hose test to watertight trunks, tunnels and ventilators.

Regulation 17
Internal watertight integrity of passenger ships above the bulkhead deck

hose test to watertight trunks, tunnels and ventilators.

Regulation 17
Internal watertight integrity of passenger ships above the bulkhead deck

1 The Administration may require that all reasonable and practicable measures shall be taken to limit the entry and spread of water above the bulkhead deck. Such measures may include partial bulkheads or webs. When partial watertight bulkheads and webs are

fitted on the bulkhead deck, above or in the i mmediate vicinity of watertight bulkheads, they shall have watertight shell and bulkhead de ck connections so as to restrict the flow of water along the deck when the ship is in a heeled damaged condition. Where the partial watertight bulkhead does not line up with the bulkhea d below, the bulkhead deck

of water along the deck when the ship is in a heeled damaged condition. Where the partial watertight bulkhead does not line up with the bulkhea d below, the bulkhead deck between shall be made effec tively watertight. Where openi ngs, pipes, scuppers, electric cables etc. are carried through the partial watertight bulkheads or decks within the immersed part of the bulkhead deck, arrangemen ts shall be made to ensure the watertight integrity of the structure above the bulkhead deck.

2 All openings in the exposed weather deck shall have coamings of ample height and strength and shall be provided with effi cient means for expeditiously closing them weathertight. Freeing ports, open rails and sc uppers shall be fitted as necessary for rapidly clearing the weather deck of water under all weather conditions.

3 The open end of air pipes terminating within a superstructure shall be at least 1 m

rapidly clearing the weather deck of water under all weather conditions.

3 The open end of air pipes terminating within a superstructure shall be at least 1 m above the waterline when the ship heels to an angle of 15º, or the maximum angle of heel during intermediate stages of flooding, as de termined by direct calculation, whichever is the greater. Alternatively, air pipes from tanks other than oil tanks may discharge through the side of the superstructure . The provisions of this paragraph are without prejudice to the provisions of the International Convention on Load Lines in force.

4 Sidescuttles, gangway, cargo and fuelli ng ports and other means for closing openings in the shell plating above the bulkhead deck shall be of efficient design and construction and of sufficient strength having regard to the spaces in which they are fitted and their positions relative to the deepest subdivision draught.

construction and of sufficient strength having regard to the spaces in which they are fitted and their positions relative to the deepest subdivision draught.

5 Efficient inside deadlights, so arranged that they can be easily and effectively closed and secured watertight, shall be provided for all side scuttles to spaces below the first deck above the bulkhead deck.

Regulation 17-1
Integrity of the hull and superstructure, damage prevention and control on ro-ro passenger ships

1.1 Subject to the provisions of paragraphs 1.2 and 1.3, all accesses that lead to spaces below the bulkhead deck shall have a lowest point which is not less than 2.5 m above the bulkhead deck.

1.2 Where vehicle ramps are installed to give access to spaces below the bulkhead deck, their openings shall be ab le to be closed weathertight to prevent ingress of water below, alarmed and indicated to the navigation bridge.

deck, their openings shall be ab le to be closed weathertight to prevent ingress of water below, alarmed and indicated to the navigation bridge.

1.3 The Administration may permit the fitting of particular accesses to spaces below the bulkhead deck provided they are necessary for the essential working of the ship, e.g., the movement of machinery and stor es, subject to such accesses being made watertight, alarmed and indicated on the navigation bridge.

2 Indicators shall be provided on the navi gation bridge for all shell doors, loading doors and other closing appliances which, if left open or not properly secured, could, in the opinion of the Administration, lead to flooding of a special category space or ro-ro space. The indicator system shall be designed on the fail-safe principle and shall show by visual alarms if the door is not fully closed or if any of the securing arrangements are not

space. The indicator system shall be designed on the fail-safe principle and shall show by visual alarms if the door is not fully closed or if any of the securing arrangements are not in place and fully locked and by audible alarms if such door or closing appliances become

open or the securing arrangements become un secured. The indicator panel on the navigation bridge shall be equipped with a mode selection function harbour/sea voyage so arranged that an audible alarm is given on the navigation bridge if the ship leaves harbour with the bow doors, inner doors, stern ramp or any other side shell doors not closed or any closing device not in the co rrect position. The power supply for the indicator system shall be independent of the power supply for operating and securing the doors.

closed or any closing device not in the co rrect position. The power supply for the indicator system shall be independent of the power supply for operating and securing the doors.

3 Television surveillance and a water leakag e detection system shall be arranged to provide an indication to the navigation bridge and to the en gine control station of any leakage through inner and outer bow doors, stern doors or any other shell doors which could lead to flooding of special category spaces or ro-ro spaces.

PART B-3
SUBDIVISION LOAD LINE ASSIGNMENT FOR PASSENGER SHIPS

Regulation 18
Assigning, marking and recording of subdivision load lines for passenger ships

1 In order that the required degree of s ubdivision shall be maintained, a load line corresponding to the approved s ubdivision draught shall be assigned and marked on the ships sides. A ship intended for alternat ing modes of operation may, if the owners

corresponding to the approved s ubdivision draught shall be assigned and marked on the ships sides. A ship intended for alternat ing modes of operation may, if the owners desire, have one or more additional load lines assigned and marked to correspond with the subdivision draughts which the Administration may approve fo r the alternative service configurations. Each service configuration so approved shall comply with part B-1 of this chapter independently of the results obtained for other modes of operation.

2 The subdivision load lines assigned and marked shall be recorded in the Passenger Ship Safety Certificate, and shall be dist inguished by the notation P1 for the principal passenger service configuration, and P2, P3, et c., for the alternative configurations. The principal passenger configuration shall be ta ken as the mode of operation in which the required subdivision index R will have the highest value.

principal passenger configuration shall be ta ken as the mode of operation in which the required subdivision index R will have the highest value.

3 The freeboard corresponding to each of these load lines shall be measured at the same position and from the same deck line as the freeboards determined in accordance with the International Convention on Load Lines in force.

4 The freeboard corresponding to each approved subdivision load line and the service configuration, for which it is approved, shall be clearly indicated on the Passenger Ship Safety Certificate.

5 In no case shall any subdivision load lin e mark be placed a bove the deepest load line in salt water as determined by the strength of the ship or the International Convention on Load Lines in force.

6 Whatever may be the position of the subdivi sion load line marks, a ship shall in no case be loaded so as to submerge the load line mark appropriate to the season and locality

6 Whatever may be the position of the subdivi sion load line marks, a ship shall in no case be loaded so as to submerge the load line mark appropriate to the season and locality as determined in accordance with the International Convention on Load Lines in force.

7 A ship shall in no case be so loaded that when it is in salt water the subdivision load line mark appropriate to the partic ular voyage and service configuration is submerged.

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(adopted on 8 December 2006)
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PART B-4
STABILITY MANAGEMENT

Regulation 19
Damage control information

1 There shall be permanently exhibited, or readily available on the navigation bridge, for the guidance of the officer in ch arge of the ship, plans showing clearly for each deck and hold the boundaries of the wate rtight compartments, the openings therein

bridge, for the guidance of the officer in ch arge of the ship, plans showing clearly for each deck and hold the boundaries of the wate rtight compartments, the openings therein with the means of closure and position of any controls thereof, and the arrangements for the correction of any list due to flooding. In addition, booklets containing the aforementioned information shall be made available to the officers of the ship.

2 Watertight doors in passenger ships permitted to remain open during navigation shall be clearly indicated in the ship's stability information.

3 General precautions to be included shal l consist of a listing of equipment, conditions, and operational procedures, considered by the Administration to be necessary to maintain watertight integrity under normal ship operations.

4 Specific precautions to be included shall consist of a listing of elements (i.e. closures, security of cargo, sounding of alarms, etc.) considered by the

4 Specific precautions to be included shall consist of a listing of elements (i.e. closures, security of cargo, sounding of alarms, etc.) considered by the Administration to be vital to the survival of the ship, passengers and crew.

5 In case of ships to which damage stability requirements of part B-1 apply, damage stability information shall provide the mast er a simple and easily understandable way of assessing the ships survivability in all damage cases involving a compartment or group of compartments.

Regulation 20
Loading of passenger ships

1 On completion of loading of the ship and prior to its departure, the master shall determine the ships trim and stability and also ascertain and record that the ship is in compliance with stability criteria in relevant regulations. The determination of the ships stability shall always be made by calculation. The Administration may accept the use of

compliance with stability criteria in relevant regulations. The determination of the ships stability shall always be made by calculation. The Administration may accept the use of an electronic loading and stability computer or equivalent means for this purpose.

2 Water ballast should not in general be car ried in tanks intended for oil fuel. In ships in which it is not practicable to avoid putting water in oil fuel tanks, oily-water separating equipment to the satisfaction of th e Administration shall be fitted, or other alternative means, such as discharge to shor e facilities, acceptable to the Administration shall be provided for disposing of the oily-water ballast.

3 The provisions of this regulation are w ithout prejudice to th e provisions of the International Convention for the Prevention of Pollution from Ships in force.
RESOLUTION MSC.216(82)
(adopted on 8 December 2006)

International Convention for the Prevention of Pollution from Ships in force.
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Regulation 21
Periodical operation and inspection of watertight doors, etc., in passenger ships

1 Drills for the operating of watertight doors, sidescuttles, valves and closing mechanisms of scuppers, ash-chutes and rubbish-chutes shall take place weekly. In ships in which the voyage exceeds one week in duration a complete drill shall be held before leaving port, and others thereafter at least once a week during the voyage.

2 All watertight doors, both hinged and power-operated, in watertight bulkheads, in use at sea, shall be operated daily.

3 The watertight doors and all mechanisms and indicators conn ected therewith, all valves, the closing of which is necessary to make a compartment watertight, and all

3 The watertight doors and all mechanisms and indicators conn ected therewith, all valves, the closing of which is necessary to make a compartment watertight, and all valves the operation of which is necessary fo r damage control cross connections shall be periodically inspected at sea at least once a week.

4 A record of all drills and inspections re quired by this regulation shall be entered in the log-book with an explicit record of any defects which may be disclosed.

Regulation 22
Prevention and control of water ingress, etc.

1 All watertight doors shall be kept closed during navi gation except that they may be opened during navigation as sp ecified in paragraphs 3 and 4. Watertight doors of a width of more than 1.2 m in machinery sp aces as permitted by regulation 13.10 may only be opened in the circumstances detailed in th at regulation. Any door which is opened in accordance with this paragraph shall be ready to be immediately closed.

be opened in the circumstances detailed in th at regulation. Any door which is opened in accordance with this paragraph shall be ready to be immediately closed.

2 Watertight doors located below the bulkhead deck having a maximum clear opening width of more than 1.2 m shall be kept closed when the ship is at sea, except for limited periods when absolutely necessary as determined by the Administration.

3 A watertight door may be opened duri ng navigation to permit the passage of passengers or crew, or when work in the im mediate vicinity of the door necessitates it being opened. The door must be immediately closed when transit through the door is complete or when the task which necessitated it being open is finished.

4 Certain watertight doors may be permitted to remain open during navigation only if considered absolutely necessary; that is, be ing open is determined essential to the safe

4 Certain watertight doors may be permitted to remain open during navigation only if considered absolutely necessary; that is, be ing open is determined essential to the safe and effective operation of the ships m achinery or to permit passengers normally unrestricted access throughout the passenger area. Such determination shall be made by the Administration only after careful consideration of the impact on ship operations and survivability. A watertight door permitted to re main thus open shall be clearly indicated in the ships stability information and shall always be ready to be immediately closed.

5 Portable plates on bulkheads shall always be in place before the ship leaves port, and shall not be removed during navigation except in case of urgent necessity at the discretion of the master. The necessary precautions shall be taken in replacing them to ensure that the joints are watertight. Power-operated sliding watertight doors permitted in

discretion of the master. The necessary precautions shall be taken in replacing them to ensure that the joints are watertight. Power-operated sliding watertight doors permitted in machinery spaces in accordance with regulati on 13.10 shall be closed before the ship leaves port and shall remain closed during navigation except in case of urgent necessity at the discretion of the master.

6 Watertight doors fitted in watertight bulkheads divi ding cargo between deck spaces in accordance with regulation 13.9.1 shall be closed before the voyage commences and shall be kept closed during navigation; the time of opening such doors in port and of closing them before the ship leaves port shall be entered in the logbook.

7 Gangway, cargo and fuelling ports fitted below the bulkhead deck shall be

closing them before the ship leaves port shall be entered in the logbook.

7 Gangway, cargo and fuelling ports fitted below the bulkhead deck shall be effectively closed and secured watertight befo re the ship leaves por t, and shall be kept closed during navigation.

8 The following doors, located above the bulkhead deck, sh all be closed and locked before the ship proceeds on any voyage and shal l remain closed and locked until the ship is at its next berth:

.1 cargo loading doors in the shel l or the boundaries of enclosed superstructures;

.2 bow visors fitted in positions as indicated in paragraph 8.1;

.3 cargo loading doors in th e collision bulkhead; and

.4 ramps forming an alternative closure to those defined in paragraphs 8.1 to 8.3 inclusive.

9 Provided that where a door cannot be opened or closed while the ship is at the berth such a door may be opened or left open while the ship appro aches or draws away

to 8.3 inclusive.

9 Provided that where a door cannot be opened or closed while the ship is at the berth such a door may be opened or left open while the ship appro aches or draws away from the berth, but only so far as may be n ecessary to enable the door to be immediately operated. In any case, the inner bow door must be kept closed.

10 Notwithstanding the requirements of para graphs 8.1 and 8.4, the Administration may authorize that particular doors can be op ened at the discretion of the master, if necessary for the operation of the ship or the embarking and disembarking of passengers when the ship is at safe anchorage and provided that the safety of the ship is not impaired.

11 The master shall ensure that an effec tive system of supervision and reporting of the closing and opening of the doors referred to in paragraph 8 is implemented.

12 The master shall ensure, before the ship proceeds on any voyage, that an entry in

the closing and opening of the doors referred to in paragraph 8 is implemented.

12 The master shall ensure, before the ship proceeds on any voyage, that an entry in the log-book is made of the time of the last closing of the doors specified in paragraph 13 and the time of any opening of particular doors in accordance with paragraph 14.

13 Hinged doors, portable plates, sidescut tles, gangway, cargo and bunkering ports and other openings, which are required by these regulations to be kept closed during navigation, shall be closed before the ship leaves port. The time of closing and the time of opening (if permissible under these regulation s) shall be recorded in such log-book as may be prescribed by the Administration.

14 Where in a between-decks, the sills of a ny of the sidescuttles referred to in regulation 15.3.2 are below a line drawn parallel to the bulkhead deck at side and having

14 Where in a between-decks, the sills of a ny of the sidescuttles referred to in regulation 15.3.2 are below a line drawn parallel to the bulkhead deck at side and having its lowest point 1.4 m plus 2.5% of the breadth of the ship above the water when the ship departs from any port, all the sidescuttles in that between-decks shall be closed watertight and locked before the ship leaves port, a nd they shall not be ope ned before the ship

arrives at the next port. In the application of this paragraph the appropriate allowance for fresh water may be made when applicable.

.1 The time of opening such sidescuttles in port and of cl osing and locking them before the ship leaves port shall be entered in such log-book as may be prescribed by the Administration.

.1 The time of opening such sidescuttles in port and of cl osing and locking them before the ship leaves port shall be entered in such log-book as may be prescribed by the Administration.

.2 For any ship that has one or more sidescuttles so placed that the requirements of paragraph 14 would apply when it was floating at its deepest subdivision draught, the Admi nistration may indicate the limiting mean draught at which these sidescuttles will have their sills above the line drawn parallel to the bul khead deck at side, and having its lowest point 1.4 m plus 2.5% of the breadth of the ship above the waterline corresponding to the limiting mean dr aught, and at which it will therefore be permissible to depart from port w ithout previously closing and locking them and to open them at sea on the re sponsibility of the master during the voyage to the next port. In tropical zones as defined in the International

them and to open them at sea on the re sponsibility of the master during the voyage to the next port. In tropical zones as defined in the International Convention on Load Lines in force, this limiting draught may be increased by 0.3 m.

15 Sidescuttles and their deadlights which will not be accessible during navigation shall be closed and secured before the ship leaves port.

16 If cargo is carried in spaces referred to in regulation 15.5.2, the sidescuttles and their deadlights shall be closed watertight and locked before the cargo is shipped and such closing and locking shall be recorded in such log-book as may be prescribed by the Administration.

17 When a rubbish-chute, etc., is not in us e, both the cover and the valve required by regulation 15.10.2 shall be kept closed and secured.

Regulation 22-1
Flooding detection systems for passenger ships carrying 36 or more persons constructed on or after 1 July 2010

regulation 15.10.2 shall be kept closed and secured.

Regulation 22-1
Flooding detection systems for passenger ships carrying 36 or more persons constructed on or after 1 July 2010

A flooding detection system for watertight spaces below the bulkhead deck shall be provided based on the guidelines developed by the Organization.

Regulation 23
Special requirements for ro-ro passenger ships

1 Special category spaces and ro-ro spaces shall be continuously patrolled or monitored by effective means, such as televi sion surveillance, so that any movement of vehicles in adverse weather conditions and unauthorized access by passengers thereto can be detected whilst the ship is underway.

2 Documented operating procedures for clos ing and securing all shell doors, loading doors and other closing appliances which, if left open or not properly secured, could, in the opinion of the Administration, lead to flooding of a special category space or ro-ro

doors and other closing appliances which, if left open or not properly secured, could, in the opinion of the Administration, lead to flooding of a special category space or ro-ro space, shall be kept on board and posted at an appropriate place.

3 All accesses from the ro-ro deck and vehi cle ramps that lead to spaces below the bulkhead deck shall be closed before the ship leaves the berth on any voyage and shall remain closed until the ship is at its next berth.

4 The master shall ensure that an effec tive system of supervision and reporting of the closing and opening of such accesses referred to in paragraph 3 is implemented.

5 The master shall ensure, before the ship leaves the berth on any voyage, that an entry in the log-book, as requir ed by regulation 22.13, is made of the time of the last

5 The master shall ensure, before the ship leaves the berth on any voyage, that an entry in the log-book, as requir ed by regulation 22.13, is made of the time of the last closing of the accesses referred to in paragraph 3.

6 Notwithstanding the requirements of para graph 3, the Administration may permit some accesses to be opened during the voyage, but only for a period sufficient to permit through passage and, if required, for the essential working of the ship.

7 All transverse or longitudinal bulkheads which are taken into account as effective to confine the seawater accumulated on the ro-ro deck shall be in place and secured before the ship leaves the berth and remain in place and secured until the ship is at its next berth.

8 Notwithstanding the requirements of para graph 7, the Administration may permit some accesses within such bulkheads to be opened during the voyage but only for

berth.

8 Notwithstanding the requirements of para graph 7, the Administration may permit some accesses within such bulkheads to be opened during the voyage but only for sufficient time to permit through passage and, if required, for the essential working of the ship.

9 In all ro-ro passenger ships, the master or the designated officer shall ensure that, without the expressed consent of the master or the designated officer, no passengers are allowed access to an enclosed ro-ro deck when the ship is under way.

Regulation 24
Prevention and control of water ingress, etc., in cargo ships

1 Openings in the shell plati ng below the deck limiting the vertical extent of damage shall be kept permanently closed while at sea.

2 Notwithstanding the requirements of paragraph 3, the Administration may authorize that particular doors may be opened at the discretion of the master, if necessary

2 Notwithstanding the requirements of paragraph 3, the Administration may authorize that particular doors may be opened at the discretion of the master, if necessary for the operation of the ship and provided that the safety of the ship is not impaired.

3 Watertight doors or ramps fitted to inte rnally subdivide large cargo spaces shall be closed before the voyage commences and shall be kept closed during navigation; the time of opening such doors in port and of closing them before th e ship leaves port shall be entered in the logbook.

4 The use of access doors and hatch covers intended to ensure the watertight integrity of internal openings shall be authorized by the officer of the watch.

Regulation 25
Water level detectors on single hold cargo ships other than bulk carriers

1 Single hold cargo ships other than bulk carriers constructed before 1 January 2007 shall comply with the requirements of this regulation not later than 31 December 2009.

1 Single hold cargo ships other than bulk carriers constructed before 1 January 2007 shall comply with the requirements of this regulation not later than 31 December 2009.

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2 Ships having a length (L) of less than 80 m, or 100 m if constructed before 1 July 1998, and a single cargo hold below the freeboard deck or cargo holds below the freeboard deck which are not separated by at least one bulkhead made watertight up to that deck, shall be fitte d in such space or spaces with water level detectors.

3 The water level detectors re quired by paragraph 2 shall:

.1 give an audible and visual alarm at the navigation bridge when the water level above the inner bottom in the cargo hold reaches a height of not less than 0.3 m, and another when such level reaches not more than 15% of the mean depth of the cargo hold; and

level above the inner bottom in the cargo hold reaches a height of not less than 0.3 m, and another when such level reaches not more than 15% of the mean depth of the cargo hold; and

.2 be fitted at the aft e nd of the hold, or above it s lowest part where the inner bottom is not parallel to the designed waterline. Where webs or partial watertight bulkheads are fitted above the inner bottom, Administrations may require the fitting of additional detectors.

4 The water level detectors required by pa ragraph 2 need not be fitted in ships complying with regulation XII/ 12, or in ships having water tight side compartments each side of the cargo hold length extending vert ically at least from inner bottom to freeboard deck.

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ANNEX 3

AMENDMENTS TO THE INTERNATIONAL CONVENTION FOR

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AMENDMENTS TO THE INTERNATIONAL CONVENTION FOR THE SAFETY OF LIFE AT SEA, 1974, AS AMENDED- 53 -

ANNEX 3

AMENDMENTS TO THE INTERNATIONAL CONVENTION FOR
THE SAFETY OF LIFE AT SEA, 1974, AS AMENDED

CHAPTER II-1
CONSTRUCTION STRUCTURE, SUBDIVISION AND STABILITY,
MACHINERY AND ELECTRICAL INSTALLATIONS

PART D
ELECTRICAL INSTALLATIONS

Regulation 41 Main source of electrical power and lighting systems

1 The following new paragraph 6 is a dded after the existing paragraph 5:

6 In passenger ships, supp lementary lighting shall be provided in all cabins to clearly indicate the exit so that occupants will be able to find their way to the door. Such lighting, which may be connected to an emergency source of power or have a self-contained source of elect rical power in each cabin, shall automatically illuminate when power to the normal cabin lighting is lost and remain on for a minimum of 30 min.

self-contained source of elect rical power in each cabin, shall automatically illuminate when power to the normal cabin lighting is lost and remain on for a minimum of 30 min.

2 The following new part F is added after the existing regulation 54:

PART F
ALTERNATIVE DESIGN AND ARRANGEMENTS

Regulation 55
Alternative design and arrangements

1 Purpose

The purpose of this regulation is to provide a methodology for alternative design and arrangements for machinery and electrical installations.

2 General

2.1 Machinery and electrical installation design and arrangements may deviate from the requirements set out in parts C, D and E, provided that the al ternative design and arrangements meet the intent of the requirements concerned and provide an equivalent level of safety to this chapter.

2.2 When alternative design or arrangements deviate from the prescriptive requirements of parts C, D and E, an engine ering analysis, evaluation and approval of the

2.2 When alternative design or arrangements deviate from the prescriptive requirements of parts C, D and E, an engine ering analysis, evaluation and approval of the design and arrangements shall be carried out in accordance with this regulation.

3 Engineering analysis

The engineering analysis shall be prepared and submitted to the Administration, based on the guidelines developed by the Organizati on and shall include, as a minimum, the following elements:

.1 determination of the ship type, machinery, electrical installations and space(s) concerned;

.2 identification of the prescriptive re quirement(s) with which the machinery and electrical installations will not comply;

.3 identification of the reason th e proposed design will not meet the

and electrical installations will not comply;

.3 identification of the reason th e proposed design will not meet the prescriptive requirements supported by compliance with other recognized engineering or industry standards;

.4 determination of the performance criteria for the ship, machinery, electrical installation or the space(s) concerned a ddressed by the relevant prescriptive requirement(s):

.1 performance criteria shall provide a level of safety not inferior to the relevant prescriptive require ments contained in parts C, D and E; and

.2 performance criteria shall be quantifiable and measurable;

.5 detailed description of the alternat ive design and arrangements, including a list of the assumptions used in the design and any proposed operational restrictions or conditions;

.6 technical justification demonstrat ing that the alternative design and arrangements meet the safety performance criteria; and

restrictions or conditions;

.6 technical justification demonstrat ing that the alternative design and arrangements meet the safety performance criteria; and

.7 risk assessment based on identifica tion of the potential faults and hazards associated with the proposal.

4 Evaluation of the alternative design and arrangements

4.1 The engineering analysis required in paragraph 3 shal l be evaluated and approved by the Administration, taking into account the guidelines developed by the Organization.

4.2 A copy of the documentation, as approve d by the Administration, indicating that the alternative design and arrangements comply with this regulation, shall be carried on board the ship.

5 Exchange of information

The Administration shall communicate to the Organization pertinent information concerning alternative design and arrangements approved by them for circulation to all Contracting Governments.

6 Re-evaluation due to change of conditions

concerning alternative design and arrangements approved by them for circulation to all Contracting Governments.

6 Re-evaluation due to change of conditions

If the assumptions and operation al restrictions that were s tipulated in th e alternative design and arrangements are changed, the engi neering analysis shall be carried out under the changed condition and shall be approved by the Administration.

RESOLUTION MSC.216(82)
(adopted on 8 December 2006)
AMENDMENTS TO THE INTERNATIONAL CONVENTION FOR THE SAFETY OF LIFE AT SEA, 1974, AS AMENDED- 55 -

CHAPTER II-2
CONSTRUCTION FIRE PROTECTION, FIRE DETECTION AND FIRE EXTINCTION

Regulation 3 Definitions

3 The following new paragraphs 51 and 52 ar e added after the existing paragraph 50:

51 Safe area in the context of a casualty is, from the perspective of habitability, any area(s) which is not flooded or which is outside the main vertical zone(s) in which a fire

51 Safe area in the context of a casualty is, from the perspective of habitability, any area(s) which is not flooded or which is outside the main vertical zone(s) in which a fire has occurred such that it can safely accomm odate all persons onboard to protect them from hazards to life or health and provide them with basic services.

52 Safety centre is a control station dedicated to the management of emergency situations. Safety systems operation, control and/or monitoring are an integral part of the safety centre.

Regulation 7 Detection and alarm

4 The following new paragraph 2.4 is a dded after the existing paragraph 2.3:

2.4 A fixed fire detection an d fire alarm system for passenger ships shall be capable of remotely and individually identifying each detector and manually operated call point.

5 In paragraphs 5.2 and 5.3.1, the following new text is added at the end of the paragraphs:

remotely and individually identifying each detector and manually operated call point.

5 In paragraphs 5.2 and 5.3.1, the following new text is added at the end of the paragraphs:

Detectors fitted in cabins, when activated, shall also be capable of emitting, or cause to be emitted, an audible alarm within the space where they are located.

Regulation 8 Control of smoke spread

6 In paragraph 2, the following new sentence is added at the end of the paragraph:

The ventilation system servi ng safety centres may be derived from the ventilation system serving the navigation bridge, unless located in an adjacent main vertical zone.

Regulation 9 Containment of fire

7 In paragraph 2.2.3.2.2 (7), the words Sale shops are deleted.

8 In paragraph 2.2.3.2.2 (8), the words Sale shops are added.

9 In the notes for tables 9.3 and 9.4, th e following sentence is added at the end of subscript c:

8 In paragraph 2.2.3.2.2 (8), the words Sale shops are added.

9 In the notes for tables 9.3 and 9.4, th e following sentence is added at the end of subscript c:

No fire rating is required for those partiti ons separating the navigation bridge and the safety centre when the latter is within the navigation bridge.

RESOLUTION MSC.216(82)
(adopted on 8 December 2006)
AMENDMENTS TO THE INTERNATIONAL CONVENTION FOR THE SAFETY OF LIFE AT SEA, 1974, AS AMENDED- 56 -

10 The following new paragraph 2.2.7 is added after paragraph 2.2.6:

2.2.7 Protection of atriums

2.2.7.1 Atriums shall be within enclosures formed of A class divisions having a fire rating determined in accordance with tables 9.2 and 9.4, as applicable.

2.2.7.2 Decks separating spaces within atrium s shall have a fire rating determined in accordance with tables 9.2 and 9.4, as applicable.

11 The existing paragraph 7.5.1 is renumbered as paragraph 7.5.1.1 and the following new

accordance with tables 9.2 and 9.4, as applicable.

11 The existing paragraph 7.5.1 is renumbered as paragraph 7.5.1.1 and the following new paragraph 7.5.1.2 is added thereafter:

7.5.1.2 Exhaust ducts from ranges for cooking equipment installed on open decks shall conform to paragraph 7.5.1.1, as a pplicable, when passing through accommodation spaces or spaces containing combustible materials.

12 The following new paragraph 7.6 is adde d after the existing paragraph 7.5.2.1:

7.6 Ventilation systems for main laundries in ships carrying more than 36 passengers

Exhaust ducts from main laundries shall be fitted with:

.1 filters readily removable for cleaning purposes;

.2 a fire damper located in the lower end of the duct which is automatically and remotely operated;

.3 remote-control arrangements for shutting off the exhaust fans and supply fans from within the space and for op erating the fire damper mentioned in paragraph 7.6.2; and

.3 remote-control arrangements for shutting off the exhaust fans and supply fans from within the space and for op erating the fire damper mentioned in paragraph 7.6.2; and

.4 suitably located hatches for inspection and cleaning.

Regulation 10 Fire fighting

13 In the first sentence of paragraph 6.4, be tween the words equipment and shall, the words installed in enclosed spaces or on open decks are added.

Regulation 13 Means of escape

14 In paragraph 3.2.3, the words public spaces in the third sentence are deleted and the following new sentence is added before the fourth sentence:

Public spaces may also have direct access to stairway enclosures except for the backstage of a theatre.

15 The following new paragraph 3.2.5.3 is added after the existing paragraph 3.2.5.2:

3.2.5.3 In lieu of the escape route ligh ting system required by paragraph 3.2.5.1, alternative evacuation guidance system s may be accepted if approved by the

3.2.5.3 In lieu of the escape route ligh ting system required by paragraph 3.2.5.1, alternative evacuation guidance system s may be accepted if approved by the Administration based on the guidelines developed by the Organization.

16 The following new regulations 21, 22 and 23 are added after the existing regulation 20:
RESOLUTION MSC.216(82)
(adopted on 8 December 2006)
AMENDMENTS TO THE INTERNATIONAL CONVENTION FOR THE SAFETY OF LIFE AT SEA, 1974, AS AMENDED- 57 -

Regulation 21
Casualty threshold, safe return to port and safe areas

1 Application

Passenger ships constructed on or after 1 July 2010 having a length, as defined in regulation II-1/2.5, of 120 m or more or having three or more main vertical zones shall comply with the provisions of this regulation.

2 Purpose

The purpose of this regulation is to establish design criteria for a ships safe return to port

comply with the provisions of this regulation.

2 Purpose

The purpose of this regulation is to establish design criteria for a ships safe return to port under its own propulsion after a casualty that does not exceed the casualty threshold stipulated in paragraph 3 and also provi des functional requireme nts and performance standards for safe areas.

3 Casualty threshold

The casualty threshold, in the context of a fire, includes:

.1 loss of space of origin up to th e nearest A class boundaries, which may be a part of the space of origin, if the space of origin is protected by a fixed fire-extinguishing system; or

.2 loss of the space of origin and adja cent spaces up to the nearest A class boundaries, which are not part of the space of origin.

4 Safe return to port

When fire damage does not exceed the casualty threshold indicated in paragraph 3, the ship shall be capable of returning to port while providing a safe area as defined in

4 Safe return to port

When fire damage does not exceed the casualty threshold indicated in paragraph 3, the ship shall be capable of returning to port while providing a safe area as defined in regulation 3.51. To be deemed capable of returning to port, the following systems shall remain operational in the remaining part of the ship not affected by fire:

.1 propulsion;

.2 steering systems and steering-control systems;

.3 navigational systems;

.4 systems for fill, transfer and service of fuel oil;

.5 internal communication between the bridge, engineering spaces, safety centre, fire-fighting and damage cont rol teams, and as required for passenger and crew notification and mustering;

.6 external communication;

.7 fire main system;

.8 fixed fire-extinguishing systems;

.8 fixed fire-extinguishing systems;

.9 fire and smoke detection system;

.10 bilge and ballast system;

.11 power-operated watertight and semi-watertight doors;

.12 systems intended to support safe areas as indicated in paragraph 5.1.2;

.13 flooding detection systems; and

.14 other systems determined by the Ad ministration to be vital to damage control efforts.

5 Safe area(s)

5.1 Functional requirements:

.1 the safe area(s) shall generally be an internal space(s); however, the use of an external space as a safe area may be allowed by the Administration taking into account any restriction due to the area of operation and relevant expected environmental conditions;

.2 the safe area(s) shall provide a ll occupants with the following basic

taking into account any restriction due to the area of operation and relevant expected environmental conditions;

.2 the safe area(s) shall provide a ll occupants with the following basic services to ensure that the health of passengers and crew is maintained:

.1 sanitation;

.2 water;

.3 food;

.4 alternate space for medical care;

.5 shelter from the weather;

.6 means of preventing heat stress and hypothermia;

.7 light; and

.8 ventilation;

.3 ventilation design shall reduce the ri sk of smoke and hot gases that could affect the use of the safe area(s); and

.4 means of access to life-saving applia nces shall be provided from each area identified or used as a safe area, ta king into account that a main vertical zone may not be available for internal transit.

5.2 Alternate space for medical care

Alternate space for medical care shall co nform to a standard acceptable to the Administration.

RESOLUTION MSC.216(82)
(adopted on 8 December 2006)

5.2 Alternate space for medical care

Alternate space for medical care shall co nform to a standard acceptable to the Administration.

RESOLUTION MSC.216(82)
(adopted on 8 December 2006)
AMENDMENTS TO THE INTERNATIONAL CONVENTION FOR THE SAFETY OF LIFE AT SEA, 1974, AS AMENDED- 59 -

Regulation 22
Design criteria for systems to remain operational after a fire casualty

1 Application

Passenger ships constructed on or after 1 July 2010 having a length, as defined in regulation II-1/2.2, of 120 m or more or having three or more main vertical zones shall comply with the provisions of this regulation.

2 Purpose

The purpose of this regulation is to provide desi gn criteria for systems required to remain operational for supporting the orderly evacua tion and abandonment of a ship, if the casualty threshold, as defined in regulation 21.3, is exceeded.

3 Systems

3.1 In case any one main vertical zone is unserviceable due to fire, the following

casualty threshold, as defined in regulation 21.3, is exceeded.

3 Systems

3.1 In case any one main vertical zone is unserviceable due to fire, the following systems shall be so arranged and segregated as to remain operational:

.1 fire main;

.2 internal communications (in support of fire-f ighting as required for passenger and crew notification and evacuation);

.3 means of external communications;

.4 bilge systems for removal of fire-fighting water;

.5 lighting along escape routes, at as sembly stations and at embarkation stations of life-saving appliances; and

.6 guidance systems for evacuation shall be available.

3.2 The above systems shall be capable of operation for at least 3 h based on the assumption of no damage outside the unservic eable main vertical zone. These systems are not required to remain operational within the unserviceable main vertical zones.

3.3 Cabling and piping within a trunk constr ucted to an A-60 standard shall be

are not required to remain operational within the unserviceable main vertical zones.

3.3 Cabling and piping within a trunk constr ucted to an A-60 standard shall be deemed to remain intact and serviceable while passing through the unserviceable main vertical zone for the purposes of paragraph 3. 1. An equivalent de gree of protection for cabling and piping may be approved by the Administration.

Regulation 23
Safety centre on passenger ships

1 Application

Passenger ships constructed on or after 1 July 2010 shall have on board a safety centre complying with the requirements of this regulation.

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(adopted on 8 December 2006)
AMENDMENTS TO THE INTERNATIONAL CONVENTION FOR THE SAFETY OF LIFE AT SEA, 1974, AS AMENDED- 60 -

2 Purpose

The purpose of this regulation is to provide a space to assist with the management of emergency situations.

3 Location and arrangement

2 Purpose

The purpose of this regulation is to provide a space to assist with the management of emergency situations.

3 Location and arrangement

The safety centre shall either be a part of the navigation bridge or be located in a separate space adjacent, but having direct access, to the navigation bridge, so that the management of emergencies can be performed without distracting watch officers from their navigational duties.

4 Layout and ergonomic design

The layout and ergonomic design of the sa fety centre shall take into account the guidelines developed by the Organization, as appropriate.

5 Communications

Means of communication between the safety centre, the central control station, the navigation bridge, the engine control room, the storage room(s) for fire-extinguishing system(s) and fire equipment lockers shall be provided.

6 Control and monitoring of safety systems

navigation bridge, the engine control room, the storage room(s) for fire-extinguishing system(s) and fire equipment lockers shall be provided.

6 Control and monitoring of safety systems

Notwithstanding the requirements set out elsewhere in the Convention, the full functionality (operation, contro l, monitoring or any combinati on thereof, as required) of the safety systems listed below shall be available from the safety centre:

.1 all powered ventilation systems;

.2 fire doors;

.3 general emergency alarm system;

.4 public address system;

.5 electrically powered ev acuation guidance systems;

.6 watertight and semi-watertight doors;

.7 indicators for shell doors, loadi ng doors and other closing appliances;

.8 water leakage of inner/outer bow doors, stern doors and any other shell door;

.9 television surveillance system;

.10 fire detection and alarm system;

.11 fixed fire-fighting local application system(s);

shell door;

.9 television surveillance system;

.10 fire detection and alarm system;

.11 fixed fire-fighting local application system(s);

.12 sprinkler and equivalent systems;

.13 water-based fire-extinguishi ng systems for machinery spaces;

.14 alarm to summon the crew;

.15 atrium smoke extraction system;

.16 flooding detection systems; and

.17 fire pumps and emergency fire pumps.

CHAPTER III
LIFE-SAVING APPLIANCES AND ARRANGEMENTS

Regulation 4 Evaluation, testing and approval of life-saving appliances and arrangements

17 Paragraph 3 is replaced by the following:

3 Before giving approval to novel life-s aving appliances or arrangements, the Administration shall ensure that such:

.1 appliances provide safety standard s at least equivalent to the requirements

Administration shall ensure that such:

.1 appliances provide safety standard s at least equivalent to the requirements of this chapter and the Code and have been evaluated and tested based on the guidelines developed by the Organization; or

.2 arrangements have successfully undergone an engi neering analysis, evaluation and approval in accordance with regulation 38.

18 The following new part C is added after the existing regulation 37:

PART C
ALTERNATIVE DESIGN AND ARRANGEMENTS

Regulation 38
Alternative design and arrangements

1 Purpose

The purpose of this regulation is to provide a methodology for alternative design and arrangements for life-saving appliances and arrangements.

2 General

2.1 Life-saving appliances and arrangement s may deviate from the requirements set out in part B, provided that the alternative de sign and arrangements meet the intent of the

2 General

2.1 Life-saving appliances and arrangement s may deviate from the requirements set out in part B, provided that the alternative de sign and arrangements meet the intent of the requirements concerned and provide an equivalent level of safety to this chapter.

2.2 When alternative design or arrangements deviate from the prescriptive requirements of part B, an engineering anal ysis, evaluation and approval of the design and arrangements shall be carried out in accordance with this regulation.
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(adopted on 8 December 2006)
AMENDMENTS TO THE INTERNATIONAL CONVENTION FOR THE SAFETY OF LIFE AT SEA, 1974, AS AMENDED- 62 -

3 Engineering analysis

The engineering analysis shall be prepared and submitted to the Administration, based on the guidelines developed by the Organizati on and shall include, as a minimum, the following elements:

.1 determination of the ship type and the life-saving appliance and arrangements concerned;

following elements:

.1 determination of the ship type and the life-saving appliance and arrangements concerned;

.2 identification of the prescriptive requ irement(s) with which the life-saving appliance and arrangements will not comply;

.3 identification of the reason th e proposed design will not meet the prescriptive requirements supported by compliance with other recognized engineering or industry standards;

.4 determination of the performance criteria for the ship and the life-saving appliance and arrangements concer ned addressed by the relevant prescriptive requirement(s):

.4.1 performance criteria shall provide a level of safety not inferior to the relevant prescriptive requirements contained in part B; and

.4.2 performance criteria shall be quantifiable and measurable;

.5 detailed description of the alternat ive design and arrangements, including a list of the assumptions used in the design and any proposed operational

.5 detailed description of the alternat ive design and arrangements, including a list of the assumptions used in the design and any proposed operational restrictions or conditions;

.6 technical justification demonstrat ing that the alternative design and arrangements meet the safety performance criteria; and

.7 risk assessment based on identifica tion of the potential faults and hazards associated with the proposal.

4 Evaluation of the alternative design and arrangements

4.1 The engineering analysis required in paragraph 3 shal l be evaluated and approved by the Administration, taking into account the guidelines developed by the Organization.

4.2 A copy of the documentation, as approve d by the Administration, indicating that the alternative design and arrangements comply with this regulation, shall be carried on board the ship.

5 Exchange of information

The Administration shall communicate to the Organization pertinent information

board the ship.

5 Exchange of information

The Administration shall communicate to the Organization pertinent information concerning alternative design and arrangements approved by them for circulation to all Contracting Governments.

6 Re-evaluation due to change of conditions

If the assumptions and operation al restrictions that were s tipulated in th e alternative design and arrangements are changed, the engi neering analysis shall be carried out under the changed condition and shall be approved by the Administration.

RESOLUTION MSC.216(82)
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AMENDMENTS TO THE INTERNATIONAL CONVENTION FOR THE SAFETY OF LIFE AT SEA, 1974, AS AMENDEDRESOLUTION MSC.216(82)
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RESOLUTION MSC.216(82)

(adopted on 8 December 2006)

AMENDMENTS TO THE INTERNATIONAL CONVENTION FOR THE SAFETY OF LIFE AT SEA, 1974, AS AMENDED

THE MARITIME SAFETY COMMITTEE,

RECALLING Article 28(b) of the Convention on the International Maritime Organization concerning the functions of the Committee,

RECALLING FURTHER article VIII(b) of the International Convention for the Safety of Life at Sea (SOLAS), 1974 (hereinafter referred to as the Convention), concerning the amendment procedure applicable to the Annex to the Convention, other than to the provisions of chapter I thereof,

HAVING CONSIDERED, at its eighty-second session, amendments to the Convention, proposed and circulated in accordance with article VIII(b)(i) thereof,

chapter I thereof,

HAVING CONSIDERED, at its eighty-second session, amendments to the Convention, proposed and circulated in accordance with article VIII(b)(i) thereof,

1. ADOPTS, in accordance with article VIII(b)(iv) of the Convention, amendments to the Convention, the text of which is set out in Annexes 1, 2 and 3 to the present resolution;

2. DETERMINES, in accordance with article V III(b)(vi)(2)(bb) of the Convention, that:

(a) the said amendments, set out in Annex 1, shall be deemed to have been accepted on 1 January 2008;

(b) the said amendments, set out in Annex 2, shall be deemed to have been accepted on 1 July 2008; and

(c) the said amendments, set out in Annex 3, shall be deemed to have been accepted on 1 January 2010,

unless, prior to those dates, more than one third of th e Contracting Governments to the Convention or Contracting Governments the comb ined merchant fleets of which constitute not

unless, prior to those dates, more than one third of th e Contracting Governments to the Convention or Contracting Governments the comb ined merchant fleets of which constitute not less than 50% of the gross tonnage of the worlds merchant fleet, have notified their objections to the amendments;

3. INVITES SOLAS Contracting Government s to note that, in accordance with article VIII(b)(vii)(2) of the Convention:

(a) the amendments, set out in Annex 1, shall enter into force on 1 July 2008;

(b) the amendments, set out in Annex 2, sha ll enter into force on 1 January 2009; and

(c) the amendments, set out in Annex 3, shall enter into force on 1 July 2010,

upon their acceptance in accordance with paragraph 2 above;
RESOLUTION MSC.216(82)
(adopted on 8 December 2006)
AMENDMENTS TO THE INTERNATIONAL CONVENTION FOR THE SAFETY OF LIFE AT SEA, 1974, AS AMENDED- 2 -

4. REQUESTS the Secretary-General, in conformity with article VIII(b)(v) of the

4. REQUESTS the Secretary-General, in conformity with article VIII(b)(v) of the Convention, to transmit certified copies of the present resolution and the text of the amendments contained in Annexes 1, 2 and 3 to all Contracting Governments to the Convention;

5. FURTHER REQUESTS the Secret ary-General to transmit copies of this resolution and its Annexes 1, 2 and 3 to Members of the Orga nization, which are not C ontracting Governments to the Convention.

RESOLUTION MSC.216(82)
(adopted on 8 December 2006)

ANNEX 1

AMENDMENTS TO THE INTERNATIONAL CONVENTION FOR THE SAFETY OF LIFE AT SEA, 1974, AS AMENDED

CHAPTER II-1
CONSTRUCTION STRUCTURE, SUBDIVISION AND STABILITY, MACHINERY AND ELECTRICAL INSTALLATIONS

PART A-1
STRUCTURE OF SHIPS

THE SAFETY OF LIFE AT SEA, 1974, AS AMENDED

CHAPTER II-1
CONSTRUCTION STRUCTURE, SUBDIVISION AND STABILITY,
MACHINERY AND ELECTRICAL INSTALLATIONS

PART A-1
STRUCTURE OF SHIPS

Regulation 3-2 Corrosion prevention of se awater ballast tanks in oil tankers and bulk carriers

1 The existing text and the heading of re gulation 3-2 are replaced by the following:

Protective coatings of dedicated seawater ballast tanks in all types of ships and double-side skin spaces of bulk carriers

1 Paragraphs 2 and 4 of this regulation sh all apply to ships of not less than 500 gross tonnage:

.1 for which the building contract is placed on or after 1 July 2008; or

.2 in the absence of a building contract, the keels of which are laid or which are at a similar stage of construction on or after 1 January 2009; or

.3 the delivery of which is on or after 1 July 2012.

2 All dedicated seawater balla st tanks arranged in ships and double-side skin spaces

.3 the delivery of which is on or after 1 July 2012.

2 All dedicated seawater balla st tanks arranged in ships and double-side skin spaces arranged in bulk carriers of 150 m in length and upwards shall be coated during construction in accordance with the Performance standard for protective coatings for dedicated seawater ballast tanks in all types of ships and double-side skin spaces of bulk carriers, adopted by the Maritime Safety Committee by resolution MSC.215(82), as may be amended by the Organization, provided that such amendments are adopted, brought into force and take effect in accordance with the provisions of article VIII of the present Convention concerning the amendment proc edures applicable to the Annex other than chapter I.

3 All dedicated seawater ba llast tanks arranged in oil tankers and bulk carriers constructed on or after 1 July 1998, for which paragraph 2 is not applicable, shall comply

than chapter I.

3 All dedicated seawater ba llast tanks arranged in oil tankers and bulk carriers constructed on or after 1 July 1998, for which paragraph 2 is not applicable, shall comply with the requirements of regulation II-1/3-2 adopted by resolution MSC.47(66).

4 Maintenance of the protective coating sy stem shall be included in the overall ships maintenance scheme. The effectiveness of the protective coating system shall be verified during the life of a ship by the Admi nistration or an organization recognized by the Administration, based on the guidelines developed by the Organization.

RESOLUTION MSC.216(82)
(adopted on 8 December 2006)
AMENDMENTS TO THE INTERNATIONAL CONVENTION FOR THE SAFETY OF LIFE AT SEA, 1974, AS AMENDED- 4 -

CHAPTER II-2
CONSTRUCTION FIRE PROTECTION, FIRE DETECTION AND FIRE EXTINCTION

Regulation 1 Application

2 In paragraph 2.2.3, the second occurren ce of the word and is deleted.

CHAPTER II-2
CONSTRUCTION FIRE PROTECTION, FIRE DETECTION AND
FIRE EXTINCTION

Regulation 1 Application

2 In paragraph 2.2.3, the second occurren ce of the word and is deleted.

3 In paragraph 2.2.4, . is replaced by ; and.

4 In paragraph 2.2, the following new subp aragraph .5 is added after the existing subparagraph .4:

.5 regulations 5.3.1.3.2 and 5.3.4 to passenger ships not later than the date of the first survey after 1 July 2008.

Regulation 3 Definitions

5 The following new paragraph 53 is a dded after the existing paragraph 52:

53 Cabin balcony is an open deck space which is provided for the exclusive use of the occupants of a single cabin and has direct access from such a cabin.

Regulation 4 Probability of ignition

6 The following text is added at the end of paragraph 5.2.3:

except that A-0 class standard is accepta ble for windows and sidescuttles outside the limit specified in regulation 9.2.4.2.5.

6 The following text is added at the end of paragraph 5.2.3:

except that A-0 class standard is accepta ble for windows and sidescuttles outside the limit specified in regulation 9.2.4.2.5.

7 In paragraph 4.4, the words or if app lied on cabin balconies of passenger ships constructed on or after 1 July 2008, are added between the words stations and shall.

Regulation 5 Fire growth potential

8 In paragraph 3.1.2.1, the last sentence is deleted.

9 The following new paragraph 3.1.3 is inserted:

3.1.3 Partial bulkheads and decks on passenger ships

3.1.3.1 Partial bulkheads or decks used to subdivide a space for utility or artistic treatment shall be of non-combustible materials.

3.1.3.2 Linings, ceilings and partial bulkheads or decks used to screen or to separate adjacent cabin balconies shall be of non-com bustible materials. Cabin balconies on passenger ships constructed before 1 July 2008 shall comply with the requirements of this

adjacent cabin balconies shall be of non-com bustible materials. Cabin balconies on passenger ships constructed before 1 July 2008 shall comply with the requirements of this paragraph by the first survey after 1 July 2008.

RESOLUTION MSC.216(82)
(adopted on 8 December 2006)
AMENDMENTS TO THE INTERNATIONAL CONVENTION FOR THE SAFETY OF LIFE AT SEA, 1974, AS AMENDED- 5 -

10 In the first sentence of paragraph 3.2.1.1, the words and cabin balconies are added between the words spaces and which, and the following new sentence is added at the end of the paragraph:

However, the provisions of paragraph 3.2.3 need not be applied to cabin balconies.

11 The following new subparagraph .3 is a dded to the existing paragraph 3.2.4.1:

.3 exposed surfaces of cabin balconies, except for natural hard wood decking systems.

12 The following new paragraph 3.4 is a dded after the existing paragraph 3.3:

.3 exposed surfaces of cabin balconies, except for natural hard wood decking systems.

12 The following new paragraph 3.4 is a dded after the existing paragraph 3.3:

3.4 Furniture and furnishings on cabin balconies of passenger ships

On passenger ships, furniture and furnis hings on cabin balconies shall comply with regulations 3.40.1, 3.40.2, 3.40.3, 3.40.6 and 3.40.7 unless such balconies are protected by a fixed pressure water-sprayi ng and fixed fire dete ction and fire alarm systems complying with regulations 7.10 and 10.6.1.3. Passenger ships constructed before 1 July 2008 shall comply with the re quirements of this paragraph by the first survey after 1 July 2008.

Regulation 6 Smoke generation potential and toxicity

13 The existing paragraph 2 is renumbered as paragraph 2.1.

14 The following new paragraph 2.2 is adde d after the renumbered paragraph 2.1:

2.2 On passenger ships constructed on or after 1 July 2008, paints, varnishes and other

14 The following new paragraph 2.2 is adde d after the renumbered paragraph 2.1:

2.2 On passenger ships constructed on or after 1 July 2008, paints, varnishes and other finishes used on exposed surfaces of cab in balconies, excluding natural hard wood decking systems, shall not be capable of producing excessive quantities of smoke and toxic products, this being determined in accordance with the Fire Test Procedures Code.

15 The existing paragraph 3 is renumbered as paragraph 3.1.

16 The following new paragraph 3.2 is adde d after the renumbered paragraph 3.1:

3.2 On passenger ships constructed on or af ter 1 July 2008, primary deck coverings on cabin balconies shall not give rise to smoke , toxic or explosive hazards at elevated temperatures, this being determined in accordance with the Fire Test Procedures Code.

Regulation 7 Detection and alarm

17 The following new paragraph 10 is a dded after the existing paragraph 9.4:

Regulation 7 Detection and alarm

17 The following new paragraph 10 is a dded after the existing paragraph 9.4:

10 Protection of cabin balconies on passenger ships

A fixed fire detection and fire alarm syst em complying with th e provisions of the Fire Safety Systems Code shall be installed on cabin balconies of ships to which regulation 5.3.4 applies, when furniture and furnishings on such balconies are not as defined in regulations 3.40.1, 3.40.2, 3.40.3, 3.40.6 and 3.40.7.

RESOLUTION MSC.216(82)
(adopted on 8 December 2006)
AMENDMENTS TO THE INTERNATIONAL CONVENTION FOR THE SAFETY OF LIFE AT SEA, 1974, AS AMENDED- 6 -

Regulation 9 Containment of fire

18 The following new paragraph 2.2.6 is added after the existing paragraph 2.2.5.2:

2.2.6 Arrangement of cabin balconies

On passenger ships constructed on or af ter 1 July 2008, non-load bearing partial bulkheads which separate adjacent cabin balc onies shall be capable of being opened by

On passenger ships constructed on or af ter 1 July 2008, non-load bearing partial bulkheads which separate adjacent cabin balc onies shall be capable of being opened by the crew from each side for the purpose of fighting fires.

Regulation 10 Fire fighting

19 The heading of paragraph 6.1 is replaced by the following:

6.1 Sprinkler and water-spraying systems in passenger ships

20 The following new paragraph 6.1.3 is added after the existing paragraph 6.1.2:

6.1.3 A fixed pressure water-spraying fire-e xtinguishing system complying with the provisions of the Fire Safety Systems Code sh all be installed on cabin balconies of ships to which regulation 5.3.4 applies, where furnitu re and furnishings on such balconies are not as defined in regulations 3.40.1, 3.40.2, 3.40.3, 3.40.6 and 3.40.7.

CHAPTER III
LIFE-SAVING APPLIANCES AND ARRANGEMENTS

Regulation 6 Communications

21 Paragraph 4.3 is replaced by the following:

CHAPTER III
LIFE-SAVING APPLIANCES AND ARRANGEMENTS

Regulation 6 Communications

21 Paragraph 4.3 is replaced by the following:

4.3 The general emergency alarm system shall be audible throughout all the accommodation and normal crew working spaces. On passenger ships, the system shall also be audible on all open decks.

Regulation 11 Survival craft muster and embarkation arrangements

22 In the first sentence of paragraph 7, the word unfavourable is replaced by the word all and the unit ° is inserted after the terms 10 and 20.

Regulation 14 Stowage of rescue boats

23 The words , and if the inflated type, in a fully inflated condition at all times are added at the end of subparagraph .1.

Regulation 19 Emergency training and drills

24 Paragraph 3.3.4 is replaced by the following:

3.3.4 In the case of a lifeboat arranged for free-fall launching, at least once every three

Regulation 19 Emergency training and drills

24 Paragraph 3.3.4 is replaced by the following:

3.3.4 In the case of a lifeboat arranged for free-fall launching, at least once every three months during an abandon ship drill, the crew shall board the lifeboat, properly secure themselves in their seats and commence launch procedures up to, but not including, the actual release of the lifeboat (i .e., the release hook shall not be releas ed). The lifeboat shall then either be free-fall launched with only the required opera ting crew on board, or lowered into the water by means of the seconda ry means of launching with or without the

operating crew on board. In both cases, the lif eboat shall thereafter be manoeuvred in the water by the operating crew. At intervals of not more than six months, the lifeboat shall

operating crew on board. In both cases, the lif eboat shall thereafter be manoeuvred in the water by the operating crew. At intervals of not more than six months, the lifeboat shall either be launched by free-fall with only the operating crew on board, or simulated launching shall be carried out in accordance with the guidelines developed by the Organization.

Regulation 20 Operational readiness, maintenance and inspections

25 Paragraphs 4.1 and 4.2 are replaced by the following:

Falls used in launching shall be inspected periodically with special regard for areas passing through sheaves, and renewed when necessa ry due to deteriora tion of the falls or at intervals of not more than 5 years, whichever is the earlier.

26 In the third sentence of paragraph 6.2, the wo rds it should be run for such period as prescribed in the manufacturers handbook are repl aced by the words a suitable water supply may be provided.

prescribed in the manufacturers handbook are repl aced by the words a suitable water supply may be provided.

27 The heading of paragraph 8 is replaced by the following:

8 Servicing of inflatable liferafts, inflat able lifejackets, marine evacuation systems and maintenance and repair of inflated rescue boats

28 The second sentence of paragraph 11.1.3 is replaced by the following:

The load to be applied shall be the mass of the survival craft or rescue boat without persons on board, except that, at intervals not exceeding five years, the test shall be carried out with a proof load equal to 1.1 times the weight of the surv ival craft or rescue boat and its full complement of persons and equipment.

29 The chapeau of paragraph 11.2 is replaced by the following:

11.2 Lifeboat or rescue boat on-load releas e gear, including free- fall lifeboat release systems, shall be:

29 The chapeau of paragraph 11.2 is replaced by the following:

11.2 Lifeboat or rescue boat on-load releas e gear, including free- fall lifeboat release systems, shall be:

30 In the first sentence of paragraph 11.2.3, th e word lifeboat is replaced by the word boat.

31 The following new paragraph 11.3 is added to the regulation:

11.3 Davit-launched liferaft auto matic release hooks shall be:

.1 maintained in accordance with instru ctions for on-board maintenance as required by regulation 36;

.2 subject to a thorough examination and operational test during the annual surveys required by regulations I/7 and I/8 by properly trained personnel familiar with the system; and

.3 operationally tested under a load of 1.1 times the total mass of the liferaft when loaded with its full complement of persons and equipment whenever the automatic release hook is overhauled. Such over-hauling and test shall be carried out at least once every five years.

the automatic release hook is overhauled. Such over-hauling and test shall be carried out at least once every five years.

RESOLUTION MSC.216(82)
(adopted on 8 December 2006)
AMENDMENTS TO THE INTERNATIONAL CONVENTION FOR THE SAFETY OF LIFE AT SEA, 1974, AS AMENDED- 8 -

Regulation 21 Survival craft and rescue boats

32 The chapeau of paragraph 1.2 is replaced by the following:

1.2 Passenger ships engaged on short in ternational voyages shall carry:

33 Paragraph 1.3 is deleted and the remaini ng paragraphs are renumbered accordingly.

34 In paragraph 1.4, the words after all person s have been assembled, with lifejackets donned are added at the end of the paragraph.

35 Paragraph 2.3 is replaced by the following:

2.3 A lifeboat may be accepted as a rescue boat provided that it and its launching and recovery arrangements also comply with the requirements for a rescue boat.

2.3 A lifeboat may be accepted as a rescue boat provided that it and its launching and recovery arrangements also comply with the requirements for a rescue boat.

36 In paragraph 3.2, the words and complying with the special standards of subdivision prescribed by regulation II-1/6.5 are deleted.

Regulation 26 Additional requirements for ro-ro passenger ships

37 In paragraph 3.1, the words approved by th e Administration having regard for the recommendations approved by the Organization are replaced by the words complying with section 5.1.4 of the Code.

38 In paragraph 3.2, all the words after the wo rd appliance are replaced by the words complying with section 6.1.7 of the Code.

Regulation 31 Survival craft and rescue boats

39 Subparagraph .2 of paragraph 1.1 is replaced by the following:

.2 in addition, one or more inflatable or rigid liferafts, complying with the

39 Subparagraph .2 of paragraph 1.1 is replaced by the following:

.2 in addition, one or more inflatable or rigid liferafts, complying with the requirements of section 4.2 or 4.3 of the Code, of a mass of less than 185 kg and stowed in a position providing for easy side -to-side transfer at a single open deck level, and of such aggregate capacity as will accommodate the total number of persons on board. If the liferaft or liferafts are not of a mass of less than 185 kg and stowed in a position providing for easy side-to-side transfer at a single open deck level, the total capacity availabl e on each side shall be sufficient to accommodate the total number of persons on board.

40 Subparagraph .2 of paragraph 1.3 is replaced by the following:

.2 unless the liferafts required by paragr aph 1.3.1 are of a mass of less than 185 kg and stowed in a position providing for easy side-to-side transfer at a single open

.2 unless the liferafts required by paragr aph 1.3.1 are of a mass of less than 185 kg and stowed in a position providing for easy side-to-side transfer at a single open deck level, additional liferafts shall be provided so that the total capacity available on each side will accommodate 150% of the total number of persons on board;

41 Subparagraph .4 of paragraph 1.3 is replaced by the following:

.4 in the event of any one survival craft being lost or rendere d unserviceable, there shall be sufficient survival craft available for use on each side, including any which are of a mass of less than 185 kg and stowed in a position providing for easy side-to-side transfer at a single ope n deck level, to accommodate the total number of persons on board.

number of persons on board.
RESOLUTION MSC.216(82)
(adopted on 8 December 2006)
AMENDMENTS TO THE INTERNATIONAL CONVENTION FOR THE SAFETY OF LIFE AT SEA, 1974, AS AMENDED- 9 -

42 The second sentence of paragraph 2 is replaced by the following:

A lifeboat may be accepted as a rescue boat, provided that it and its launching and recovery arrangements also comply with the requirements for a rescue boat.

Regulation 32 Personal life-saving appliances

43 In the first sentence of paragraph 3.2, the words of an appropriate size, are inserted between the words suit and complying.

44 In paragraph 3.3, the words including remo tely located survival craft carried in accordance with regulation 31.1.4, are inserted be tween the words stowed, and additional and the words of an appropriate size are inserted between the words suits and shall.

Regulation 35 Training manual and on-board training aids

and the words of an appropriate size are inserted between the words suits and shall.

Regulation 35 Training manual and on-board training aids

45 The following new paragraph 5 is a dded after the existing paragraph 4:

5 The training manual shall be written in the working language of the ship.

RESOLUTION MSC.216(82)
(adopted on 8 December 2006)
AMENDMENTS TO THE INTERNATIONAL CONVENTION FOR THE SAFETY OF LIFE AT SEA, 1974, AS AMENDED- 10 -

CHAPTER XII
ADDITIONAL SAFETY MEASURES FOR BULK CARRIERS

Regulation 6 Structural and other requirements for bulk carriers

46 The existing paragraph 3 is deleted and the existing paragraphs 4 and 5 are renumbered as paragraphs 3 and 4.

Regulation 12 Hold, ballast and dry space water ingress alarms

47 In paragraph 1.2, the reference to regulation II-1/11 is replaced by the reference to regulation II-1/12.

Regulation 13 Availability of pumping systems

47 In paragraph 1.2, the reference to regulation II-1/11 is replaced by the reference to regulation II-1/12.

Regulation 13 Availability of pumping systems

48 In paragraph 1, the reference to regulati on II-1/11.4 is replaced by the reference to regulation II-1/12.

APPENDIX
CERTIFICATES

49 In the Passenger Ship Safety Certificate, Cargo Ship Safety Construction Certificate and Cargo Ship Safety Certificate, the phrase Date on which keel was laid or ship was at a similar stage of construction or, where applicable, date on which work for a conversion or an alteration or modification of a major character was commenced ..... is replaced by the following:

Date of build:

Date of building contract .....

Date on which keel was laid or ship was at similar stage of construction .....

Date of delivery .....

Date on which work for a conversion or an alteration or modification of a major

Date of delivery .....

Date on which work for a conversion or an alteration or modification of a major character was commenced (where applicable) .....

All applicable dates shall be completed.

Record of Equipment for the Passenger Ship Safety Certificate (Form P)

50 In the Record of Equipment for the Passenger Ship Safety Certificate (Form P), the following new item 4.2 is inserted in section 5 after item 4:

4.2 Long-range identification and tracking system,

and item 4 (Automatic identification system (AIS)) is renumbered as item 4.1.

Record of Equipment for the Cargo Ship Safety Equipment Certificate (Form E)

51 In the Record of Equipment for the Cargo Ship Safety Equipment Certificate (Form E), the following new item 4.2 is inserted in section 3 after item 4:

51 In the Record of Equipment for the Cargo Ship Safety Equipment Certificate (Form E), the following new item 4.2 is inserted in section 3 after item 4:

4.2 Long-range identification and tracking system,

and item 4 (Automatic identification system (AIS)) is renumbered as item 4.1.

Record of Equipment for the Cargo Ship Safety Certificate (Form C)

52 In the Record of Equipment for the Cargo Sh ip Safety Certificate (Form C), the following new item 4.2 is inserted in section 5 after item 4:

4.2 Long-range identification and tracking system,

and item 4 (Automatic identification system (AIS)) is renumbered as item 4.1.

Form of Safety Certificate for Nuclear Passenger Ships

53 In the table of paragraph 2.1.3, in the sectio n commencing with the words THIS IS TO CERTIFY:, the reference to regula tion II-1/13 is replaced by the reference to regulation II-1/18.

RESOLUTION MSC.216(82)
(adopted on 8 December 2006)

IS TO CERTIFY:, the reference to regula tion II-1/13 is replaced by the reference to regulation II-1/18.

RESOLUTION MSC.216(82)
(adopted on 8 December 2006)
AMENDMENTS TO THE INTERNATIONAL CONVENTION FOR THE SAFETY OF LIFE AT SEA, 1974, AS AMENDED- 12 -

ANNEX 2

AMENDMENTS TO THE INTERNATIONAL CONVENTION FOR THE SAFETY OF LIFE AT SEA, 1974, AS AMENDED

CHAPTER II-1
CONSTRUCTION STRUCTURE, SUBDIVISION AND STABILITY, MACHINERY AND ELECTRICAL INSTALLATIONS

1 The existing text of parts A, B and B-1 of the chapter is replaced by the following:

PART A
GENERAL

Regulation 1
Application

1.1 Unless expressly provided otherwise, this chapter shall apply to ships the keels of which are laid or which are at a similar stage of construction on or after 1 January 2009.

1.2 For the purpose of this chapter, the term a similar stage of construction means the stage at which:

.1 construction identifiable with a specific ship begins; and

1.2 For the purpose of this chapter, the term a similar stage of construction means the stage at which:

.1 construction identifiable with a specific ship begins; and

.2 assembly of that ship has commen ced comprising at least 50 tonnes or one per cent of the estimated mass of all structural material, whichever is less.

1.3 For the purpose of this chapter:

.1 the expression ships constructed means ships the keels of which are laid or which are at a similar stage of construction;

.2 the expression all ships means ships constructed before, on or after 1 January 2009;

.3 a cargo ship, whenever built, which is converted to a passenger ship shall be treated as a passenger ship cons tructed on the date on which such a conversion commences;

.4 the expression alterations and modifications of a major character means, in the context of cargo ship subdivisi on and stability, any modification to

conversion commences;

.4 the expression alterations and modifications of a major character means, in the context of cargo ship subdivisi on and stability, any modification to the construction which affects the level of subdivision of that ship. Where a cargo ship is subject to such modifi cation, it shall be demonstrated that the A/R ratio calculated for the ship after such modifications is not less than the A/R ratio calculated for the ship before the modification. However, in those cases where the ships A/R ratio before modification is equal to or greater than unity, it is only necessary that the ship after modification has an A value which is not less than R , calculated for the modified ship.

2 Unless expressly provided otherwise, for ships constructed before 1 January 2009,

2 Unless expressly provided otherwise, for ships constructed before 1 January 2009, the Administration shall ensu re that the requirements which are applicable under chapter II-1 of the Internat ional Convention for the Safety of Life at Sea, 1974, as amended by resolutions MSC.1(XLV) , MSC.6(48), MSC.11(55), MSC.12(56), MSC.13(57), MSC.19(58), MSC.26(60), MSC.27(61), Resolution 1 of the 1995 SOLAS Conference, MSC.47(66), MSC.57(67), MSC.65(68), MSC.69(69), MSC.99(73), MSC.134(76), MSC.151(78) and MSC.170(79) are complied with.

3 All ships which undergo repairs, alterati ons, modifications a nd outfitting related thereto shall continue to comply with at least the requirements previously applicable to these ships. Such ships, if constructed before the date on which any relevant amendments

thereto shall continue to comply with at least the requirements previously applicable to these ships. Such ships, if constructed before the date on which any relevant amendments enter into force, shall, as a rule, comply w ith the requirements for ships constructed on or after that date to at least the same extent as they did before undergoing such repairs, alterations, modifications or out fitting. Repairs, alterations and modifications of a major character and outfitting related thereto shall meet the requirements for ships constructed on or after the date on which any relevant amendments enter into force, in so far as the Administration deems reasonable and practicable.

4 The Administration of a State may, if it considers that the sheltered nature and conditions of the voyage are such as to render the application of any specific requirements of this chapter unreasonable or unnecessary, exempt from those requirements individual

conditions of the voyage are such as to render the application of any specific requirements of this chapter unreasonable or unnecessary, exempt from those requirements individual ships or classes of ships entitled to fly the flag of that State which, in the course of their voyage, do not proceed more than 20 miles from the nearest land.

5 In the case of passenger ships which are employed in special trades for the carriage of large numbers of special trade pa ssengers, such as the pilgrim trade, the Administration of the State whose flag such ship s are entitled to fly, if satisfied that it is impracticable to enforce compliance with the requirements of this chapter, may exempt such ships from those requirements, prov ided that they comply fully with the provisions of:

.1 the rules annexed to the Special Trade Passenger Ships Agreement, 1971; and

.2 the rules annexed to the Protocol on Space Requirements for Special Trade Passenger Ships, 1973.

Regulation 2

and

.2 the rules annexed to the Protocol on Space Requirements for Special Trade Passenger Ships, 1973.

Regulation 2
Definitions

For the purpose of this chapter, unless expressly provided otherwise:

1 Subdivision length (Ls) of the ship is the greatest pr ojected moulded length of that part of the ship at or below deck or decks limiting the vertical extent of flooding with the ship at the deepest subdivision draught.

2 Mid-length is the mid-point of the subdivision length of the ship.

3 Aft terminal is the aft limit of the subdivision length.

4 Forward terminal is the forward limit of the subdivision length.

5 Length (L) is the length as defined in the In ternational Convention on Load Lines in force.

6 Freeboard deck is the deck as defined in the International Convention on Load Lines in force.

6 Freeboard deck is the deck as defined in the International Convention on Load Lines in force.

7 Forward perpendicular is the forward perpendicular as defined in the International Convention on Load Lines in force.

8 Breadth (B) is the greatest moulded breadth of the ship at or below the deepest subdivision draught.

9 Draught (d) is the vertical distance from th e keel line at mid-length to the waterline in question.

10 Deepest subdivision draught (d s) is the waterline which corresponds to the summer load line draught of the ship.

11 Light service draught (d l) is the service draught co rresponding to the lightest anticipated loading and associated tankage, including, however, such ballast as may be necessary for stability and/ or immersion. Passenger ship s should include the full complement of passengers and crew on board.

12 Partial subdivision draught (d p) is the light service draught plus 60% of the

complement of passengers and crew on board.

12 Partial subdivision draught (d p) is the light service draught plus 60% of the difference between the light service draught and the deepest subdivision draught.

13 Trim is the difference between the draught forward and th e draught aft, where the draughts are measured at the forward and aft terminals respectively, disregarding any rake of keel.

14 Permeability (μ) of a space is the proportion of the immersed volume of that space which can be occupied by water.

15 Machinery spaces are spaces between the wate rtight boundaries of a space containing the main and auxiliary propulsion machinery, including boilers, generators and electric motors primarily intended for propuls ion. In the case of unusual arrangements, the Administration may define the limits of the machinery spaces.

16 Weathertight means that in any sea conditions water will not penetrate into the ship.

the Administration may define the limits of the machinery spaces.

16 Weathertight means that in any sea conditions water will not penetrate into the ship.

17 Watertight means having scantlings and arrangements capable of preventing the passage of water in any direction under the h ead of water likely to occur in intact and damaged conditions. In the damaged condition, th e head of water is to be considered in the worst situation at equilibrium, including intermediate stages of flooding.

18 Design pressure means the hydrostatic pressure for which each structure or appliance assumed watertight in the intact and damage stability calculations is designed to withstand.

19 Bulkhead deck in a passenger ship means the uppe rmost deck at any point in the subdivision length (Ls) to which the main bulkheads and the ships shell are carried watertight and the lowermost deck from which passenger and crew evacuation will not be

subdivision length (Ls) to which the main bulkheads and the ships shell are carried watertight and the lowermost deck from which passenger and crew evacuation will not be impeded by water in any stage of flooding for damage cases defined in regulation 8 and in part B-2 of this chapter. The bulkhead deck may be a stepped deck. In a cargo ship the freeboard deck may be taken as the bulkhead deck.

20 Deadweight is the difference in tonnes between the displacement of a ship in water of a specific gravity of 1.025 at the draught corresponding to the assigned summer freeboard and the lightweight of the ship.

21 Lightweight is the displacement of a ship in tonnes without cargo, fuel, lubricating oil, ballast water, fresh water and feedwater in tanks, consumable stores, and passengers and crew and their effects.

oil, ballast water, fresh water and feedwater in tanks, consumable stores, and passengers and crew and their effects.

22 Oil tanker is the oil tanker defined in regulat ion 1 of Annex I of the Protocol of 1978 relating to the International Conven tion for the Prevention of Pollution from Ships, 1973.

23 Ro-ro passenger ship means a passenger ship with ro-ro spaces or special category spaces as defined in regulation II-2/3.

24 Bulk carrier means a bulk carrier as defined in regulation XII/1.1.

25 Keel line is a line parallel to the slope of the keel passing amidships through:

.1 the top of the keel at centreline or line of intersection of the inside of shell plating with the keel if a bar keel ex tends below that line, on a ship with a metal shell; or

.2 in wood and composite ships, the distance is measured from the lower edge of the keel rabbet. When the fo rm at the lower part of the midship

metal shell; or

.2 in wood and composite ships, the distance is measured from the lower edge of the keel rabbet. When the fo rm at the lower part of the midship section is of a hollow character, or where thick garboards are fitted, the distance is measured from the point where the line of the flat of the bottom continued inward intersects the centreline amidships.

26 Amidship is at the middle of the length (L).

Regulation 3
Definitions relating to parts C, D and E

For the purpose of parts C, D and E, unless expressly provided otherwise:

1 Steering gear control system is the equipment by which orders are transmitted from the navigating bridge to the steering gear power units. Steering gear control systems comprise transmitters, receivers, hydraulic control pumps and their associated motors, motor controllers, piping and cables.

2 Main steering gear is the machinery, rudder actuat ors, steering gear, power units,

motor controllers, piping and cables.

2 Main steering gear is the machinery, rudder actuat ors, steering gear, power units, if any, and ancillary equipment and the means of applying torque to the rudder stock (e.g., tiller or quadrant) necessary for effecting movement of the rudder for the purpose of steering the ship under normal service conditions.

3 Steering gear power unit is:

.1 in the case of electric steering gear, an electric motor and its associated electrical equipment;

.2 in the case of electrohydraulic st eering gear, an electric motor and its associated electrical equipment and connected pump; or

.3 in the case of other hyd raulic steering gear, a driving engine and connected pump.

4 Auxiliary steering gear is the equipment other than any part of the main steering

.3 in the case of other hyd raulic steering gear, a driving engine and connected pump.

4 Auxiliary steering gear is the equipment other than any part of the main steering gear necessary to steer the ship in the event of failure of the main steering gear but not including the tiller, quadrant or components serving the same purpose.

5 Normal operational and habitable condition is a condition under which the ship as a whole, the machinery, services, means and aids ensuring propulsion, ability to steer, safe navigation, fire and flooding safety, in ternal and external communications and signals, means of escape, and emergency boat winches, as well as the designed comfortable conditions of habitability are in working order and functioning normally.

6 Emergency condition is a condition under which any services needed for normal operational and habitable conditions are not in working order due to failure of the main source of electrical power.

operational and habitable conditions are not in working order due to failure of the main source of electrical power.

7 Main source of electrical power is a source intended to supply electrical power to the main switchboard for distribution to all serv ices necessary for maintaining the ship in normal operational and habitable conditions.

8 Dead ship condition is the condition under which the main propulsion plant, boilers and auxiliaries are not in operation due to the absence of power.

9 Main generating station is the space in which the ma in source of electrical power is situated.

10 Main switchboard is a switchboard which is direc tly supplied by the main source of electrical power and is intended to distribute electrical energy to the ships services.

11 Emergency switchboard is a switchboard which in the event of failure of the main electrical power supply system is directly s upplied by the emergency source of electrical

11 Emergency switchboard is a switchboard which in the event of failure of the main electrical power supply system is directly s upplied by the emergency source of electrical power or the transitional source of emerge ncy power and is intended to distribute electrical energy to the emergency services.

12 Emergency source of electrical power is a source of electr ical power, intended to supply the emergency switchboard in the event of a failure of the supply from the main source of electrical power.

13 Power actuating system is the hydraulic equipment provided for supplying power to turn the rudder stock, comp rising a steering gear power unit or units, together with the associated pipes and fittings, and a rudder actuator. The power actuating systems may share common mechanical components (i.e. tiller, quadrant and rudder stock) or components serving the same purpose.

14 Maximum ahead service speed is the greatest speed which the ship is designed to

components serving the same purpose.

14 Maximum ahead service speed is the greatest speed which the ship is designed to maintain in service at sea at the deepest seagoing draught.

15 Maximum astern speed is the speed which it is estimated the ship can attain at the designed maximum astern power at the deepest seagoing draught.

16 Machinery spaces are all machinery spaces of category A and all other spaces containing propelling machinery, boilers, oil fuel units, steam and internal combustion engines, generators and major electrical machinery, oil filling stations, refrigerating, stabilizing, ventilation and ai r conditioning machinery, and sim ilar spaces, and trunks to such spaces.

17 Machinery spaces of category A are those spaces and trunks to such spaces which contain:

such spaces.

17 Machinery spaces of category A are those spaces and trunks to such spaces which contain:

.1 internal combustion machinery used for main propulsion;

.2 internal combustion machinery us ed for purposes other than main propulsion where such machinery has in the aggregate a total power output of not less than 375 kW; or

.3 any oil-fired boiler or oil fuel unit.

18 Control stations are those spaces in which the ship's radio or main navigating equipment or the emergency source of power is located or where the fire recording or fire control equipment is centralized.

19 Chemical tanker is a cargo ship constructed or adapted and used for the carriage in bulk of any liquid product listed in either:

.1 chapter 17 of the International Code for the Construction and Equipment of Ships Carrying Dangerous Chemical s in Bulk adopted by the Maritime Safety Committee by resolution MSC.4(48), hereinafter referred to as

of Ships Carrying Dangerous Chemical s in Bulk adopted by the Maritime Safety Committee by resolution MSC.4(48), hereinafter referred to as the International Bulk Chemical Code, as may be amended by the Organization; or

.2 chapter VI of the Code for the Construction and Equipment of Ships Carrying Dangerous Chemicals in Bulk adopted by the Assembly of the Organization by resolution A.212(VII), hereinafter referred to as the Bulk Chemical Code, as has been or may be amended by the Organization,

whichever is applicable.

20 Gas carrier is a cargo ship constructed or ad apted and used for the carriage in bulk of any liquefied gas or other products listed in either:

.1 chapter 19 of the International Code for the Construction and Equipment of Ships Carrying Liquefied Gases in Bulk adopted by the Maritime Safety Committee by resolution MSC.5(48), he reinafter referred to as the International Gas Carrier Code, as may be amended by the Organization; or

Committee by resolution MSC.5(48), he reinafter referred to as the International Gas Carrier Code, as may be amended by the Organization; or

.2 chapter XIX of the Code for the C onstruction and Equipment of Ships Carrying Liquefied Gases in Bulk adopted by the Organization by resolution A.328(IX), herein after referred to as the Gas Carrier Code, as has been or may be amended by the Organization,

whichever is applicable.

RESOLUTION MSC.216(82)
(adopted on 8 December 2006)
AMENDMENTS TO THE INTERNATIONAL CONVENTION FOR THE SAFETY OF LIFE AT SEA, 1974, AS AMENDED- 18 -

PART B
SUBDIVISION AND STABILITY

Regulation 4
General

1 The damage stability requirements in pa rts B-1 through B-4 shall apply to cargo ships of 80 m in length (L) and upwards and to all passenge r ships regardless of length but shall exclude those cargo ships which are shown to co mply with subdivision and

ships of 80 m in length (L) and upwards and to all passenge r ships regardless of length but shall exclude those cargo ships which are shown to co mply with subdivision and damage stability regulations in other instruments developed by the Organization.

2 The Administration may, for a particular ship or group of ships, accept alternative methodologies if it is satisfied that at least the same degree of safety as represented by these regulations is achieved. Any Admi nistration which allows such alternative methodologies shall communicate to the Organization particulars thereof.

3 Ships shall be as efficiently subdivided as is possible having regard to the nature of the service for which they are intended. The degree of subdivision shall vary with the subdivision length (Ls) of the ship and with the service, in such manner that the highest degree of subdivision corresponds with th e ships of greatest subdivision length (Ls),

subdivision length (Ls) of the ship and with the service, in such manner that the highest degree of subdivision corresponds with th e ships of greatest subdivision length (Ls), primarily engaged in the carriage of passengers.

4 Where it is proposed to fit decks, inner skins or longitudinal bulkheads of sufficient tightness to seriously restrict the flow of water, the Administration shall be satisfied that proper consideration is given to beneficial or adve rse effects of such structures in the calculations.

PART B-1
STABILITY

Regulation 5
Intact stability information

1 Every passenger ship regardless of si ze and every cargo ship having a length (L) of 24 m and upwards, shall be inclined upon its completion and the elements of its stability determined.

2 The Administration may allow the inclining test of an individual cargo ship to be dispensed with provided basic stability data are available from the inclining test of a sister

2 The Administration may allow the inclining test of an individual cargo ship to be dispensed with provided basic stability data are available from the inclining test of a sister ship and it is shown to the satisfaction of the Administration that reliable stability information for the exempted ship can be obtai ned from such basic data, as required by regulation 5-1. A weight survey shall be carried out upon completion and the ship shall be inclined whenever in comparison with the data derived from the sister ship, a deviation from the lightship displacement exceeding 1% for ships of 160 m or more in length and 2% for ships of 50 m or less in length a nd as determined by linear interpolation for intermediate lengths or a deviation from th e lightship longitudinal centre of gravity exceeding 0.5% of Ls is found.

3 The Administration may also allow the inclin ing test of an individual ship or class

exceeding 0.5% of Ls is found.

3 The Administration may also allow the inclin ing test of an individual ship or class of ships especially designed for the carriage of liquids or ore in bulk to be dispensed with when reference to existing data for similar sh ips clearly indicates th at due to the ships proportions and arrangements more than sufficient metacentric height will be available in all probable loading conditions.

4 Where any alterations are made to a ship so as to materially affect the stability information supplied to the master, amended stability information shall be provided. If necessary the ship shall be re-inclined. The ship shall be re-inclined if anticipated deviations exceed one of the values specified in paragraph 5.

necessary the ship shall be re-inclined. The ship shall be re-inclined if anticipated deviations exceed one of the values specified in paragraph 5.

5 At periodical intervals not exceeding five years, a lightweight survey shall be carried out on all passenger ships to verify any changes in lightship displacement and longitudinal centre of gravity. The ship shall be re-inclined whenever, in comparison with the approved stability information, a de viation from the lightship displacement exceeding 2% or a deviation of the longitu dinal centre of gravity exceeding 1% of Ls is found or anticipated.

6 Every ship shall have scales of draughts marked clearly at the bow and stern. In the case where the draught mark s are not located where they are easily readable, or operational constraints for a particular trade make it difficult to read the draught marks, then the ship shall also be fitted with a re liable draught indicatin g system by which the

operational constraints for a particular trade make it difficult to read the draught marks, then the ship shall also be fitted with a re liable draught indicatin g system by which the bow and stern draughts can be determined.

Regulation 5-1
Stability information to be supplied to the master

1 The master shall be supplied with such information satisfactory to the Administration as is necessary to enable hi m by rapid and simple processes to obtain accurate guidance as to the st ability of the ship under vary ing conditions of service. A copy of the stability information shall be furnished to the Administration.

2 The information should include:

.1 curves or tables of minimum operational metacentric height (GM) versus draught which assures compliance with the relevant intact and damage stability requirements, alternatively co rresponding curves or tables of the maximum allowable vertical centre of gravity (KG) versus draught, or with

stability requirements, alternatively co rresponding curves or tables of the maximum allowable vertical centre of gravity (KG) versus draught, or with the equivalents of either of these curves;

.2 instructions concerning the operation of cross-flooding arrangements; and

.3 all other data and aids which might be necessary to maintain the required intact stability and stability after damage.

3 The stability information shall show the influence of various trims in cases where the operational trim range exceeds +/- 0.5% of Ls.

4 For ships which have to fulfil the stabil ity requirements of part B-1, information referred to in paragraph 2 are determined fr om considerations related to the subdivision index, in the following manner: Minimum required GM (or maximum permissible vertical position of centre of gravity KG) for the three draughts ds, dp and dl are equal to the GM (or KG values) of corresponding loading cases used for the calculation of survival

vertical position of centre of gravity KG) for the three draughts ds, dp and dl are equal to the GM (or KG values) of corresponding loading cases used for the calculation of survival factor si. For intermediate draughts, values to be used shall be obtained by linear interpolation applied to the GM value only between the deep est subdivision draught and the partial subdivision draught and between the pa rtial load line and the light service draught respectively. Int act stability criteria will also be taken in to account by retaining for each draft the maximum among minimum required GM values or the minimum of maximum permissible KG values for both criteria. If th e subdivision index is calculated for different trims, several required GM curves will be established in the same way.

5 When curves or tables of mini mum operational metacentric height (GM) versus draught are not appropriate, the master should ensure that the operating condition does not deviate from a studied loading condition, or verify by calculation that the stability criteria are satisfied for this loading condition.

Regulation 6
Required subdivision index R

1 The subdivision of a ship is considered sufficient if the attained subdivision index A, determined in accordance with regula tion 7, is not less than the required subdivision index R calculated in accordance with this regulation and if, in addition, the partial indices As, Ap and Al are not less than 0.9 R for passenger ships and 0.5 R for cargo ships.

2 For all ships to which the damage stabil ity requirements of this chapter apply, the

partial indices As, Ap and Al are not less than 0.9 R for passenger ships and 0.5 R for cargo ships.

2 For all ships to which the damage stabil ity requirements of this chapter apply, the degree of subdivision to be provided shall be determined by the required subdivision index R, as follows:

.1 In the case of cargo ships greater than 100 m in length (Ls):

R = 1521281 s + L

.2 In the case of cargo ships not less than 80 m in length (Ls) and not greater than 100 m in length (Ls):

R =)]1 100 [1/(1 1 oo sRR L× +

where Ro is the value R as calculated in accordance with the formula in subparagraph .1.

.3 In the case of passenger ships:

R = 15,2252.55,0001 s + + N L

where:

N = N1 + 2N2
N1 = number of persons for wh om lifeboats are provided
N2 = number of persons (including officers and crew) the ship is permitted to carry in excess of N1.

.4 Where the conditions of service are such that compliance with

N2 = number of persons (including officers and crew) the ship is permitted to carry in excess of N1.

.4 Where the conditions of service are such that compliance with paragraph 2.3 of this regulation on the basis of N = N1 + 2N2 is impracticable and where the Administration considers that a suitably reduced degree of hazard exists, a lesser value of N may be taken but in no case less than N = N1 + N2.

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Regulation 7
Attained subdivision index A

1 The attained subdivision index A is obtained by the summation of the partial indices As, Ap and Al, (weighted as shown) calculated for the draughts ds, dp and dl defined in regulation 2 in accordance with the following formula:

$$A = 0.4\,As + 0.4Ap + 0.2Al$$

Each partial index is a summation of contri butions from all damage cases taken in

in regulation 2 in accordance with the following formula:

A = 0.4 As + 0.4Ap + 0.2Al

Each partial index is a summation of contri butions from all damage cases taken in consideration, using the following formula:

A = pi si

where:

i represents each compartment or group of compartments under consideration,

pi accounts for the probability that only the compartment or group of compartments under consideration may be flooded, disregarding any horizontal subdivision, as defined in regulation 7-1,

si accounts for the probability of survival after flooding the compartment or group of compartments under consideration, and includes the effect of any horizontal subdivision, as defined in regulation 7-2.

2 In the calculation of A, the level trim shall be used for the deepest subdivision draught and the partial subdivisi on draught. The actual service trim shall be used for the

2 In the calculation of A, the level trim shall be used for the deepest subdivision draught and the partial subdivisi on draught. The actual service trim shall be used for the light service draught. If in any service conditio n, the trim variation in comparison with the calculated trim is greater than 0.5% of Ls, one or more additional calculations of A are to be submitted for the same draughts but different trims so that, for all service conditions, the difference in trim in compar ison with the reference trim used for one calculation will be less than 0.5% of Ls.

3 When determining the positive righting lever (GZ) of the residual stability curve, the displacement used should be that of th e intact condition. Th at is, the constant displacement method of calculation should be used.

4 The summation indicated by the above formula shall be taken over the ships subdivision length (Ls) for all cases of flooding in which a single compartment or two or

4 The summation indicated by the above formula shall be taken over the ships subdivision length (Ls) for all cases of flooding in which a single compartment or two or more adjacent compartments are involved. In the case of unsymmetrical arrangements, the calculated A value should be the mean value obt ained from calcul ations involving both sides. Alternatively, it should be take n as that corresponding to the side which evidently gives the least favourable result.

5 Wherever wing compartments are fitted, contribution to the summation indicated by the formula shall be taken for all cases of flooding in which wing compartments are involved. Additionally, cases of simultaneou s flooding of a wing compartment or group of compartments and the adjacent inboard compartment or group of compartments, but excluding damage of transverse extent greater than one half of the ship breadth B, may be

of compartments and the adjacent inboard compartment or group of compartments, but excluding damage of transverse extent greater than one half of the ship breadth B, may be added. For the purpose of this regulation, transverse extent is measured inboard from ships side, at right angle to the centreline at the level of the deepest subdivision draught.

6 In the flooding calculati ons carried out according to the regulations, only one breach of the hull and only one free surface ne ed to be assumed. The assumed vertical extent of damage is to extend from the baseline upwards to any watertight horizontal subdivision above the waterline or higher. However, if a lesser ex tent of damage will give a more severe result, such extent is to be assumed.

subdivision above the waterline or higher. However, if a lesser ex tent of damage will give a more severe result, such extent is to be assumed.

7 If pipes, ducts or tunnels are situated within the assumed extent of damage, arrangements are to be made to ensure that progressive flooding cannot thereby extend to compartments other than those assumed flooded. However, the Administration may permit minor progressive flooding if it is dem onstrated that its effects can be easily controlled and the safety of the ship is not impaired.

Regulation 7-1
Calculation of the factor pi

1 The factor pi for a compartment or group of comp artments shall be calculated in accordance with paragraphs 1.1 and 1.2 using the following notations:

j = the aftmost damage zone number involved in the damage starting with No.1 at the stern;

n = the number of adjacent damage zones involved in the damage;

k = is the number of a partic ular longitudinal bulkhead as barrier for transverse

No.1 at the stern;

n = the number of adjacent damage zones involved in the damage;

k = is the number of a partic ular longitudinal bulkhead as barrier for transverse penetration in a damage zone counted from shell towards the centre line. The shell has k = 0;

x1 = the distance from the aft terminal of Ls to the aft end of the zone in question;

x2 = the distance from the aft terminal of Ls to the forward end of the zone in question;

b = the mean transverse distance in metr es measured at right angles to the centreline at the deepest subdivision loadline between the shell and an assumed vertical plane extended betw een the longitudinal limits used in calculating the factor pi and which is a tangent t o, or common with, all or part of the outermost portion of the longitudinal bulkhead under consideration. This verti cal plane shall be so orientated that the mean transverse distance to the shell is a maximum, but not more than twice the

consideration. This verti cal plane shall be so orientated that the mean transverse distance to the shell is a maximum, but not more than twice the least distance between the plane and th e shell. If the upper part of a longitudinal bulkhead is below the deepest subdivision loadline the vertical plane used for determination of b is assumed to extend upwards to the deepest subdivision waterline. In any case, b is not to be taken greater than B/2.

If the damage involves a single zone only:

$$pi = p(x1j,x2j) \cdot [r(x1j,x2j,bk) - r(x1j,x2j,bk-1)]$$

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If the damage involves two adjacent zones:

$$pi = p(x1j,x2j+1) \cdot [r(x1j,x2j+1,bk) - r(x1j,x2j+1,bk-1)] - p(x1j,x2j) \cdot [r(x1j,x2j,bk) - r(x1j,x2j,bk-1)] - p(x1j+1,x2j+1) \cdot [r(x1j+1,x2j+1,bk) - r(x1j+1,x2j+1,bk-1)]$$

If the damage involves three or more adjacent zones:

- p(x1j,x2j) · [r(x1j,x2j,bk) - r(x1j,x2j,bk-1)]
- p(x1j+1,x2j+1) · [r(x1j+1,x2j+1,bk) - r(x1j+1,x2j+1,bk-1)]

If the damage involves three or more adjacent zones:

pi = p(x1j,x2j+n-1) · [r(x1j,x2j+n-1,bk) - r(x1j,x2j+n-1,bk-1)]
- p(x1j,x2j+n-2) · [r(x1j,x2j+n-2,bk) - r(x1j,x2j+n-2,bk-1)]
- p(x1j+1,x2j+n-1) · [r(x1j+1,x2j+n-1,bk) - r(x1j+1,x2j+n-1,bk-1)]
+ p(x1j+1,x2j+n-2) · [r(x1j+1,x2j+n-2,bk) - r(x1j+1,x2j+n-2,bk-1)]

and where r(x1, x2, b0) = 0

1.1 The factor p(x1, x2) is to be calculated according to the following formulae:

Overall normalized max damage length: Jmax = 10/33
Knuckle point in the distribution: Jkn = 5/33
Cumulative probability at Jkn: pk = 11/12
Maximum absolute damage length: lmax = 60 m
Length where normalized distribution ends: L* = 260 m

Probability density at J = 0:

= knkknk J JpJpb max0 12

When :

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Probability density at J = 0:

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() 211 214 kkk k mk JpJ J Jpb =

()
221
12
k m
k
J J
pb

=

mJ b b2122 =

The non-dimensional damage length:

()
sL
1 x 2 xJ =
The normalized length of a compartment or group of compartments:

Jn is to be taken as the lesser of J and Jm

1.1.1 Where neither limits of the comp artment or group of compartments under

The normalized length of a compartment or group of compartments:

Jn is to be taken as the lesser of J and Jm

1.1.1 Where neither limits of the comp artment or group of compartments under consideration coincides with the aft or forward terminals:

J Jk :
() 121121 3b J b J61) , (+ = =p 2 x 1 xp

J>Jk :
() ()
3 3
2112
2
1211
3
112
3
1
2
1
3
1) , (k nkkk J J b JJ b J b J b J b p 2 x 1 x p + + = =
() ( ) () k nk n J J J b J J b J b + +22
2 2
2221
2
1

1.1.2 Where the aft limit of the compar tment or group of compartments under consideration coincides with the aft terminal or the forward limit of the compartment or group of compartments under consideration coincides with the forward terminal:

J Jk:
() J p 2 x 1 x p+ =1
2
1) , (
J>Jk:
() J p 2 x 1 x p+ =2
2
1) , (

J>Jk:
() J p 2 x 1 x p+ =2
2
1) , (

1.1.3 Where the compartment or groups of compartments considered extends over the entire subdivision length (Ls):

p(x1, x2) = 1

1.2 The factor r(x1, x2, b) shall be determined by the following formulae:

=) , (1) 1 (1) , , (2 x 1 x pGCb 2 x 1 x r

where:

() 4 45 12+ = bb J J C , where

B
bJb
= 15

1.2.1 Where the compartment or groups of compartments considered extends over the entire subdivision length (Ls):

bb J b J b G G12
2
111
2
1 + = =

1.2.2 Where neither limits of the comp artment or group of compartments under consideration coincides with the aft or forward terminals:

() 0 12
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1 JJ b J b J b J b G G+ + = = , where
() bJ J J, min0 =

consideration coincides with the aft or forward terminals:

() 0 12
2
0 1211
3
0 112
2
1
3
1 JJ b J b J b J b G G+ + = = , where
() bJ J J, min0 =

1.2.3 Where the aft limit of the compar tment or group of compartments under consideration coincides with the aft terminal or the forward limit of the compartment or group of compartments under consideration coincides with the forward terminal:

() J G G G + =1 2
2
1

Regulation 7-2
Calculation of the factor si

1 The factor si shall be determined for each case of assumed flooding, involving a compartment or group of compartments, in accordance with the following notations and the provisions in this regulation.

e is the equilibrium heel angle in any stage of flooding, in degrees;

v is the angle, in any stage of fl ooding, where the righting lever becomes negative, or the angle at which an opening incapable of being closed weathertight becomes submerged;

v is the angle, in any stage of fl ooding, where the righting lever becomes negative, or the angle at which an opening incapable of being closed weathertight becomes submerged;

GZmax is the maximum positive righting lever, in metres, up to the angle v;

Range is the range of positive righting levers, in degrees, measured from the angle e. The positive range is to be taken up to the angle v;

Flooding stage is any discrete step during the flooding process, including the stage before equalization (if any) until final equilibrium has been reached.

1.1 The factor si, for any damage case at any initial loading condition, di, shall be obtained from the formula:

$$s_i = \text{minimum} \{ s_{intermediate,i} \text{ or } s_{final,i} \cdot s_{mom,i} \}$$

where:

1.1 The factor si, for any damage case at any initial loading condition, di, shall be obtained from the formula:

si = minimum { sintermediate,i or sfinal,i · smom,i }

where:

sintermediate,i is the probability to survive all intermediate flooding stages until the final equilibrium stage, and is calculated in accordance with paragraph 2;

sfinal,i is the probability to survive in the fina l equilibrium stage of flooding. It is calculated in accordance with paragraph 3;
1.
smom,i is the probability to survive heel ing moments, and is calculated in accordance with paragraph 4.

2 The factor sintermediate,i is applicable only to passenger ships (for cargo ships sintermediate,i should be taken as unity) and shall be taken as the least of the s-factors obtained from all flooding stages including the stage before equalization, if any, and is to be calculated as follows:

sintermediate,i =
4
1
max
7 05 . 0

RangeGZ

obtained from all flooding stages including the stage before equalization, if any, and is to be calculated as follows:

sintermediate,i =
4
1
max
7 05 . 0

RangeGZ

where GZmax is not to be taken as more than 0.05 m and Range a s n o t m o r e t h a n 7 ° . sintermediate = 0, if the intermediate heel angle exceeds 15º. Where cross-flooding fittings are required, the time for equalization shall not exceed 10 min.

3 The factor sfinal,i shall be obtained from the formula:

sfinal,i =
4
1
max
16 12 . 0

RangeGZK

where:

GZmax is not to be taken as more than 0.12 m;

Range is not to be taken as more than 16°;

K = 1 if e min

K = 0 if e max

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K =
minmax
max

e otherwise,

where:

min is 7° for passenger ships and 25° for cargo ships; and

K =
minmax
max

e otherwise,

where:

min is 7° for passenger ships and 25° for cargo ships; and

max is 15° for passenger ships and 30° for cargo ships.

4 The factor smom,i is applicable only to passenger ships (for cargo ships smom,i shall be taken as unity) and shall be calculated at the final equilibrium from the formula:

()
heel
max
mom,
M
ntDisplaceme04 . 0 GZs i =
where:

Displacement is the intact displacement at the subdivision draught;

Mheel is the maximum assumed heeling moment as calculated in accordance with paragraph 4.1; and

smom,i 1

4.1 The heeling moment Mheel is to be calculated as follows:

Mheel = maximum { Mpassenger or Mwind or MSurvivalcraft}

4.1.1 Mpassenger is the maximum assumed heeling moment resulting from movement of passengers, and is to be obtained as follows:

Mpassenger = (0.075 · Np) · (0.45 · B) (tm)

where:

4.1.1 Mpassenger is the maximum assumed heeling moment resulting from movement of passengers, and is to be obtained as follows:

Mpassenger = (0.075 · Np) · (0.45 · B) (tm)

where:

Np is the maximum number of passengers pe rmitted to be on board in the service condition corresponding to the deepest subdivision draught under consideration; and

B is the beam of the ship.

Alternatively, the heeling moment may be calculated assuming the passengers are distributed with 4 persons per square metre on available deck areas towards one side of the ship on the decks where muster stations are located and in such a way that they produce the most adverse heeling moment. In doing so, a weight of 75 kg per passenger is to be assumed.

4.1.2 Mwind is the maximum assumed wind force acting in a damage situation:

Mwind = (P · A · Z) / 9,806 (tm)

is to be assumed.

4.1.2 Mwind is the maximum assumed wind force acting in a damage situation:

Mwind = (P · A · Z) / 9,806 (tm)

where:

P = 120 N/m 2;

A = projected lateral area above waterline;

Z = distance from centre of lateral projected area above waterline to T/2; and

T = ships draught, di.

4.1.3 MSurvivalcraft is the maximum assumed heeling moment due to the launching of all fully loaded davit-launched survival craft on one side of the ship. It shall be calculated using the following assumptions:

.1 all lifeboats and rescue boats fitted on the side to which the ship has heeled after having sustained damage sha ll be assumed to be swung out fully loaded and ready for lowering;

.2 for lifeboats which are arranged to be launched fully loaded from the

after having sustained damage sha ll be assumed to be swung out fully loaded and ready for lowering;

.2 for lifeboats which are arranged to be launched fully loaded from the stowed position, the maximum heeli ng moment during launching shall be taken;

.3 a fully loaded davit-launched liferaft attached to each davit on the side to which the ship has heeled after having sustained damage shall be assumed to be swung out ready for lowering;

.4 persons not in the life-saving appl iances which are swung out shall not provide either additional heeling or righting moment; and

.5 life-saving appliances on the side of the ship opposite to the side to which the ship has heeled shall be assumed to be in a stowed position.

5 Unsymmetrical flooding is to be kept to a minimum consistent with the efficient arrangements. Where it is necessary to corr ect large angles of heel, the means adopted

5 Unsymmetrical flooding is to be kept to a minimum consistent with the efficient arrangements. Where it is necessary to corr ect large angles of heel, the means adopted shall, where practicable, be self-acting, but in any case where controls to equalization devices are provided they shall be operable from above the bulkhead deck. These fittings together with their controls shall be a cceptable to the Administration. Suitable information concerning the use of equalization devices shall be supplie d to the master of the ship.

5.1 Tanks and compartments taking part in such equalization shall be fitted with air pipes or equivalent means of sufficient cross-se ction to ensure that the flow of water into the equalization compartments is not delayed.

5.2 In all cases, si is to be taken as zero in those cases where the final waterline, taking into account sinkage, heel and trim, immerses:

.1 the lower edge of openings through which progressive flooding may take

into account sinkage, heel and trim, immerses:

.1 the lower edge of openings through which progressive flooding may take place and such flooding is not accounted for in the calculation of factor si. Such openings shall include air-pipes, ventilators and openings which are closed by means of weathertight doors or hatch covers; and

.2 any part of the bulkhead deck in passenger ships considered a horizontal evacuation route for compliance with chapter II-2.

5.3 The factor si is to be taken as zero if, taking into account sinkage, heel and trim, any of the following occur in any intermediate stage or in the final stage of flooding:

.1 immersion of any vertic al escape hatch in the bulkhead deck intended for compliance with chapter II-2;

.1 immersion of any vertic al escape hatch in the bulkhead deck intended for compliance with chapter II-2;

.2 any controls intended for the opera tion of watertight doors, equalization devices, valves on piping or on ventilation ducts intended to maintain the integrity of watertight bulkheads fr om above the bulkhead deck become inaccessible or inoperable;

.3 immersion of any part of piping or ventilation ducts carried through a watertight boundary that is located within any compartment included in damage cases contributing to the attained index A, if not fitted with watertight means of closure at each boundary.

5.4 However, where compartments assume d flooded due to prog ressive flooding are taken into account in the damage stability calculations multiple values of sintermediate,i may be calculated assuming equalization in additional flooding phases.

5.5 Except as provided in paragraph 5.3.1, openings closed by means of watertight

be calculated assuming equalization in additional flooding phases.

5.5 Except as provided in paragraph 5.3.1, openings closed by means of watertight manhole covers and flush scuttles, small wate rtight hatch covers, remotely operated sliding watertight doors, side scuttles of the non-opening type as well as watertight access doors and hatch covers required to be kept closed at sea need not be considered.

6 Where horizontal watertight boundaries are fitted above the waterline under consideration the s-value calculated for the lower compartment or group of compartments shall be obtained by multiplying the value as determined in paragraph 1.1 by the reduction factor vm according to paragraph 6.1, which represents the probability that the spaces above the horizontal subdivision will not be flooded.

6.1 The factor vm shall be obtained from the formula:

vm = v(Hj, n, m, d) v(Hj, n, m-1, d)

where:

spaces above the horizontal subdivision will not be flooded.

6.1 The factor vm shall be obtained from the formula:

vm = v(Hj, n, m, d) v(Hj, n, m-1, d)

where:

Hj, n, m is the least height above the baselin e, in metres, within the longitudinal range of x1(j)...x2(j+n-1) of the mth horizontal boundary which is assumed to limit the vertical extent of flooding for the damaged compartments under consideration;

Hj, n, m-1 is the least height above the baselin e, in metres, within the longitudinal range of x1(j)...x2(j+n-1) of the (m-1)th horizontal boundary which is assumed to limit the vertical extent of flooding for the damaged compartments under consideration;

j signifies the aft terminal of the damaged compartments under consideration;

m represents each horizontal boundary counted upwards from the waterline under consideration;
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m represents each horizontal boundary counted upwards from the waterline under consideration;

d is the draught in question as defined in regulation 2; and

x1 and x 2 represent the terminals of the co mpartment or group of compartments considered in regulation 7-1.

6.1.1 The factors v(Hj, n, m, d) and v(Hj, n, m-1, d) shall be obtained from the formulae:

, if (Hm-d) is less than, or equal to, 7.8 m;

in all other cases,

where:

v(Hj, n, m, d) is to be taken as 1, if Hm coincides with the uppermost watertight boundary of the ship within the range (x1(j)...x2(j+n-1)), and

v(Hj, n, 0, d) is to be taken as 0.

In no case is vm to be taken as less than zero or more than 1.

6.2 In general, each contribution dA to the index A in the case of horizontal

v(Hj, n, 0, d) is to be taken as 0.

In no case is vm to be taken as less than zero or more than 1.

6.2 In general, each contribution dA to the index A in the case of horizontal subdivisions is obtained from the formula:

() ( )[] mmi s vs v v s v p dA min 12 min 1 2 1 min 11 +++ =

where:

vm = the v-value calculated in accordance with paragraph 6.1;
smin = the least s-factor for all combinations of damages obtained when the assumed damage extends from the assumed damage height Hm downwards.

Regulation 7-3
Permeability

1 For the purpose of the subdivision and damage stability calculations of the regulations, the permeability of each general compartment or part of a compartment shall be as follows:

Spaces	Permeability
Appropriated to stores	0.60
Occupied by accommodation	0.95
Occupied by machinery	0.85
Void spaces	0.95
Intended for liquids	0 or 0.951

1 Whichever results in the more severe requirement.

()

Appropriated to stores 0.60
Occupied by accommodation 0.95
Occupied by machinery 0.85
Void spaces 0.95
Intended for liquids 0 or 0.951
1 Whichever results in the more severe requirement.

()
8 . 78 . 0) , (d Hd H v =
()

+ = 7 . 4 8 . 72 . 0 8 . 0) , (d Hd H v

2 For the purpose of the subdivision and damage stability calculations of the regulations, the permeability of each cargo compartment or part of a compartment shall be as follows:

Spaces	Permeability at draught ds	Permeability at draught dp	Permeability at draught dl
Dry cargo spaces	0.70	0.80	0.95
Container spaces	0.70	0.80	0.95
Ro-ro spaces	0.90	0.90	0.95
Cargo liquids	0.70	0.80	0.95

3 Other figures for permeability may be used if substantiated by calculations.

Regulation 8

Container spaces 0.70 0.80 0.95
Ro-ro spaces 0.90 0.90 0.95
Cargo liquids 0.70 0.80 0.95

3 Other figures for permeability may be used if substantiated by calculations.

Regulation 8
Special requirements concerning passenger ship stability

1 A passenger ship intended to carry 400 or more persons shall have watertight subdivision abaft the collis ion bulkhead so that s i = 1 for the three loading conditions on which is based the calculation of the subdivision index and for a damage involving all the compartments within 0.08L measured from the forward perpendicular.

2 A passenger ship intended to carry 36 or more persons is to be capable of withstanding damage along the side shell to an extent specified in paragraph 3. Compliance with this regulation is to be achieved by demonstrating that s i, as defined in regulation 7-2, is not less than 0.9 for the three loading conditions on which is based the calculation of the subdivision index.

regulation 7-2, is not less than 0.9 for the three loading conditions on which is based the calculation of the subdivision index.

3 The damage extent to be assumed when demonstrating compliance with paragraph 2, is to be dependent on both N as defined in regulation 6, and Ls as defined in regulation 2, such that:

.1 the vertical extent of damage is to extend from the ships moulded baseline to a position up to 12.5 m above the position of the deepest subdivision draught as defined in regulation 2, unless a lesser vertical extent of damage were to give a lower value of si, in which case this reduced extent is to be used;

.2 where 400 or more persons are to be carried, a damage length of 0.03L s but not less than 3 m is to be assumed at any position along the side shell, in conjunction with a penetr ation inboard of 0.1 B but not less than 0.75 m measured inboard from the ship side, at right angle to the centreline at the level of the deepest subdivision draught;

conjunction with a penetr ation inboard of 0.1 B but not less than 0.75 m measured inboard from the ship side, at right angle to the centreline at the level of the deepest subdivision draught;

.3 where less than 400 persons are carried, damage length is to be assumed at any position along the shell side betw een transverse watertight bulkheads provided that the distance between two adjacent transverse watertight bulkheads is not less than the assume d damage length. If the distance between adjacent transverse watertight bulkheads is less than the assumed damage length, only one of these bulkheads shall be considered effective for the purpose of demonstrating compliance with paragraph 2;

.4 where 36 persons are carried, a damage length of 0.015 Ls but not less

.4 where 36 persons are carried, a damage length of 0.015 Ls but not less than 3 m is to be assumed, in co njunction with a penetration inboard of 0.05B but not less than 0.75 m; and

.5 where more than 36, but fewer than 400 persons are carried the values of damage length and penetration inboard, used in the dete rmination of the assumed extent of damage, are to be obtained by linear interpolation between the values of damage length and penetration which apply for ships carrying 36 persons and 400 persons as specified in subparagraphs .4 and .2.

Regulation 8-1
System capabilities after a flooding casualty on passenger ships

1 Application

This regulation applies to passenger ships constructed on or after 1 July 2010 to which regulation II-2/21 applies.

2 Availability of essential system s in case of flooding damage

This regulation applies to passenger ships constructed on or after 1 July 2010 to which regulation II-2/21 applies.

2 Availability of essential system s in case of flooding damage

A passenger ship shall be designed so that the systems specified in regulation II-2/21.4 remain operational when the ship is subject to flooding of any single watertight compartment.

PART B-2
SUBDIVISION, WATERTIGHT AND WEATHERTIGHT INTEGRITY

Regulation 9
Double bottoms in passenger ships and cargo ships other than tankers

1 A double bottom shall be fitted extending from the collision bulkhead to the afterpeak bulkhead, as far as this is practicable and compatible with the design and proper working of the ship.

2 Where a double bottom is required to be fitted the inner bottom shall be continued out to the ship's sides in such a manner as to protect the bottom to the turn of the bilge. Such protection will be deemed satisfactory if the inner bottom is not lower at any part

out to the ship's sides in such a manner as to protect the bottom to the turn of the bilge. Such protection will be deemed satisfactory if the inner bottom is not lower at any part than a plane parallel with the keel line and which is located not less than a vertical distance h measured from the keel line, as calculated by the formula:

h = B/20

However, in no case is the value of h to be less than 760 mm, and need not be taken as more than 2,000 mm.

3 Small wells constructed in the double bottom in connection with drainage arrangements of holds, etc., shall not extend downward more than necessary. A well extending to the outer bottom is, however, permitted at the afte r end of the shaft tunnel. Other wells (e.g., for lubricating oil under main engines) may be permitted by the Administration if satisfied that the arrange ments give protection equivalent to that afforded by a double bottom complying with this regulation. In no case shall the vertical

Administration if satisfied that the arrange ments give protection equivalent to that afforded by a double bottom complying with this regulation. In no case shall the vertical distance from the bottom of such a well to a plane coinciding with the keel line be less than 500 mm.

4 A double bottom need not be fitted in way of watertight tanks, including dry tanks of moderate size, provided the safety of the ship is not impaired in the event of bottom or side damage.

5 In the case of passenger ships to which th e provisions of regulation 1.5 apply and which are engaged on regular se rvice within the limits of a short in ternational voyage as defined in regulation III/3.22, the Administration may permit a double bottom to be dispensed with if satisfied th at the fitting of a double bottom in that part would not be

defined in regulation III/3.22, the Administration may permit a double bottom to be dispensed with if satisfied th at the fitting of a double bottom in that part would not be compatible with the design and proper working of the ship.

6 Any part of a passenger ship or a cargo ship that is not fitted with a double bottom in accordance with paragraphs 1, 4 or 5 shall be capable of withstanding bottom damages, as specified in paragraph 8, in that part of the ship.

7 In the case of unusual bottom arrangement s in a passenger ship or a cargo ship, it shall be demonstrated that the ship is capable of withstanding bottom damages as specified in paragraph 8.

8 Compliance with paragraphs 6 or 7 is to be achieved by demonstrating that si, when calculated in accordance with regulatio n 7-2, is not less than 1 for all service conditions when subject to a bottom damage assumed at any position along the ships

when calculated in accordance with regulatio n 7-2, is not less than 1 for all service conditions when subject to a bottom damage assumed at any position along the ships bottom and with an extent specified in subparagraph .2 for the affected part of the ship:

.1 Flooding of such spaces shall not render emergency power and lighting, internal communication, signals or ot her emergency devices inoperable in other parts of the ship.

.2 Assumed extent of damage shall be as follows:

	For 0.3 L from the forward perpendicular of the ship	Any other part of the ship
Longitudinal extent	1/3 L2/3 or 14.5 m, whichever is less	1/3 L2/3 or 14.5 m, whichever is less
Transverse extent	B/6 or 10 m, whichever is less	B/6 or 5 m, whichever is less
Vertical extent, measured from the keel line	B/20 or 2 m, whichever is less	B/20 or 2 m, whichever is less

.3 If any damage of a lesser extent than the maximum damage specified in

Vertical extent,
measured from
the keel line
B/20 or 2 m, whichever is less B/20 or 2 m, whichever is less

.3 If any damage of a lesser extent than the maximum damage specified in subparagraph .2 would result in a mo re severe condition, such damage should be considered.

9 In case of large lower ho lds in passenger ships, the Administration may require an increased double bottom height of not more than B /10 or 3 m, whichever is less, measured from the keel line. Alternativel y, bottom damages may be calculated for these areas, in accordance with paragraph 8, but assuming an increased vertical extent.

Regulation 10
Construction of watertight bulkheads

1 Each watertight subdivision bulkhead, whether transverse or longitudinal, shall be

Regulation 10
Construction of watertight bulkheads

1 Each watertight subdivision bulkhead, whether transverse or longitudinal, shall be constructed having scantlings as specified in regulation 2.17. In all cases, watertight subdivision bulkheads shall be capable of supporting at least the pressure due to a head of water up to the bulkhead deck.

2 Steps and recesses in watertight bulkheads shall be as strong as the bulkhead at the place where each occurs.

Regulation 11
Initial testing of watertight bulkheads, etc.

1 Testing watertight spaces not intended to hold liquids and cargo holds intended to hold ballast by filling them with water is not compulsory. When testing by filling with water is not carried out, a hose test shall be ca rried out where practicable. This test shall be carried out in the most advanced stage of the fitting out of the ship. Where a hose test is not practicable because of possible dama ge to machinery, electrical equipment

be carried out in the most advanced stage of the fitting out of the ship. Where a hose test is not practicable because of possible dama ge to machinery, electrical equipment insulation or outfitting items, it may be repl aced by a careful visual examination of welded connections, supported where deem ed necessary by means such as a dye penetrant test or an ultrasonic leak test or an equivalent test. In any case a thorough inspection of the watertight bulkheads shall be carried out.

2 The forepeak, double bottom (including duct ke els) and inner skins shall be tested with water to a head corresponding to the requirements of regulation 10.1.

3 Tanks which are intended to hold liquids, and which form part of the watertight subdivision of the ship, shall be tested for tightness and structural strength with water to a head corresponding to its design pressure. The water head is in no case to be less than the

subdivision of the ship, shall be tested for tightness and structural strength with water to a head corresponding to its design pressure. The water head is in no case to be less than the top of the air pipes or to a level of 2.4 m above the to p of the tank, whichever is the greater.

4 The tests referred to in paragraphs 2 and 3 are for the purpose of ensuring that the subdivision structural arrangements are watertight and are not to be regarded as a test of the fitness of any compartment for the storage of oil fuel or for other special purposes for which a test of a superior character may be required depending on the height to which the liquid has access in the tank or its connections.

Regulation 12
Peak and machinery space bulkheads, shaft tunnels, etc.

1 A collision bulkhead shall be fitted which shall be watertight up to the bulkhead deck. This bulkhead shall be located at a distance from the forward perpendicular of not

1 A collision bulkhead shall be fitted which shall be watertight up to the bulkhead deck. This bulkhead shall be located at a distance from the forward perpendicular of not less than 0.05L or 10 m, whichever is the less, and, except as may be permitted by the Administration, not more than 0.08L or 0.05L + 3 m, whichever is the greater.

2 Where any part of the ship below the waterline extends forward of the forward perpendicular, e.g., a bulbous bow, the distan ces stipulated in paragraph 1 shall be measured from a point either:

.1 at the mid-length of such extension;

.2 at a distance 0.015 L forward of the forward perpendicular; or

.3 at a distance 3 m forward of the forward perpendicular,

whichever gives the smallest measurement.

.3 at a distance 3 m forward of the forward perpendicular,

whichever gives the smallest measurement.

3 The bulkhead may have steps or recesses provided they are within the limits prescribed in paragraph 1 or 2.

4 No doors, manholes, access openings, ventilation ducts or any other openings shall be fitted in the collision bulkhead below the bulkhead deck.

5.1 Except as provided in paragraph 5.2, the collision bulkhead may be pierced below the bulkhead deck by not more than one pipe fo r dealing with fluid in the forepeak tank, provided that the pipe is fitted with a scre w-down valve capable of being operated from above the bulkhead deck, the valve chest be ing secured inside th e forepeak to the collision bulkhead. The Admini stration may, however, authorize the fitting of this valve on the after side of the collision bulkhead pr ovided that the valve is readily accessible

collision bulkhead. The Admini stration may, however, authorize the fitting of this valve on the after side of the collision bulkhead pr ovided that the valve is readily accessible under all service conditions and the space in wh ich it is located is not a cargo space. All valves shall be of steel, bron ze or other approved ductile ma terial. Valves of ordinary cast iron or similar material are not acceptable.

5.2 If the forepeak is divided to hold two different kinds of liquids the Administration may allow the collision bulkhead to be pier ced below the bulkhead deck by two pipes, each of which is fitted as required by paragraph 5.1, provided the Administration is satisfied that there is no practical alternative to the fitting of such a second pipe and that, having regard to the additional subdivision provided in the forepeak, the safety of the ship is maintained.

6 Where a long forward superstructure is fitted the collision bulkhead shall be

having regard to the additional subdivision provided in the forepeak, the safety of the ship is maintained.

6 Where a long forward superstructure is fitted the collision bulkhead shall be extended weathertight to the deck next above the bulkhead deck. The extension need not be fitted directly above the bulkhead below provided it is located within the limits prescribed in paragraph 1 or 2 with the exception permitted by paragraph 7 and that the part of the deck which forms the step is ma de effectively weathertight. The extension shall be so arranged as to preclude the possibility of the bow door causing damage to it in the case of damage to, or detachment of, a bow door.

7 Where bow doors are fitted and a sloping lo ading ramp forms part of the extension of the collision bulkhead above the bulkhead deck the ramp shall be weathertight over its complete length. In cargo ships the part of the ramp which is more than 2.3 m above the

of the collision bulkhead above the bulkhead deck the ramp shall be weathertight over its complete length. In cargo ships the part of the ramp which is more than 2.3 m above the bulkhead deck may extend forward of the limit specified in paragraph 1 or 2. Ramps not meeting the above requirements shall be disr egarded as an extension of the collision bulkhead.

8 The number of openings in the extensi on of the collision bulkhead above the freeboard deck shall be restricted to the mi nimum compatible with the design and normal operation of the ship. All such openings shall be capable of being closed weathertight.

9 Bulkheads shall be fitted separating the machinery space from cargo and accommodation spaces forward and aft and made watertight up to the bulkhead deck. In passenger ships an afterpeak bulkhead shall also be fitted and made watertight up to the bulkhead deck. The afterpeak bulkhead may, however, be stepped below the bulkhead

passenger ships an afterpeak bulkhead shall also be fitted and made watertight up to the bulkhead deck. The afterpeak bulkhead may, however, be stepped below the bulkhead deck, provided the degree of safety of the sh ip as regards subdivision is not thereby diminished.

10 In all cases stern tubes shall be enclosed in watertight spaces of moderate volume. In passenger ships the stern gland shall be situ ated in a watertight shaft tunnel or other watertight space separate from the stern tube compartment and of such volume that, if flooded by leakage through the stern gland, the bulkhead deck will not be immersed. In cargo ships other measures to minimize the dange r of water penetrating into the ship in case of damage to stern tube arrangement s may be taken at the discretion of the Administration.

Regulation 13

case of damage to stern tube arrangement s may be taken at the discretion of the Administration.

Regulation 13
Openings in watertight bulkheads below the bulkhead deck in passenger ships

1 The number of openings in watertight bul kheads shall be reduced to the minimum compatible with the design and proper working of the ship, satisfactory means shall be provided for closing these openings.

2.1 Where pipes, scuppers, electric cables , etc., are carried through watertight bulkheads, arrangements shall be made to ensure the watertight integrity of the bulkheads.

2.2 Valves not forming part of a piping sy stem shall not be permitted in watertight bulkheads.

2.3 Lead or other heat sensitive materials sh all not be used in systems which penetrate watertight bulkheads, where deterioration of such systems in the event of fire would impair the watertight integrity of the bulkheads.

3 No doors, manholes, or access openings ar e permitted in watertight transverse

impair the watertight integrity of the bulkheads.

3 No doors, manholes, or access openings ar e permitted in watertight transverse bulkheads dividing a cargo space from an adjoining cargo space, except as provided in paragraph 9.1 and in regulation 14.

4 Subject to paragraph 10, not more than one door, apart from the doors to shaft tunnels, may be fitted in each watertight bu lkhead within spaces containing the main and auxiliary propulsion machinery including boilers serving the needs of propulsion. Where two or more shafts are fitted, the tunnels shall be connected by an intercommunicating passage. There shall be only one door between the machinery space and the tunnel spaces where two shafts are fitted and only two doors where there are more th an two shafts. All these doors shall be of the sliding type and shall be so located as to have their sills as high as practicable. The hand gear for operat ing these doors from above the bulkhead deck

these doors shall be of the sliding type and shall be so located as to have their sills as high as practicable. The hand gear for operat ing these doors from above the bulkhead deck shall be situated outside the spaces containing the machinery.

5.1 Watertight doors, except as provided in paragraph 9.1 or regulation 14, shall be power-operated sliding doors complying with the requirements of paragraph 7 capable of being closed simultaneously from the central operating console at the navigation bridge in not more than 60 s with the ship in the upright position.

5.2 The means of operation whether by power or by hand of any power-operated sliding watertight door shall be capable of closing the door with the ship listed to 15o either way. Consideration shall also be given to the forces which may act on either side of the door as may be experienced when water is flowing through the opening

to 15o either way. Consideration shall also be given to the forces which may act on either side of the door as may be experienced when water is flowing through the opening applying a static head equivalent to a water height of at least 1 m above the sill on the centreline of the door.

5.3 Watertight door controls, including hydraulic piping and electric cables, shall be kept as close as practicable to the bulkhead in which the doors are fitted, in order to minimize the likelihood of them being invol ved in any damage which the ship may sustain. The positioning of watertight doors and their controls shall be such that if the ship sustains damage within one fifth of the breadth of the ship, as defined in regulation 2, such distance being measured at right angles to the centreline at the level of the deepest

ship sustains damage within one fifth of the breadth of the ship, as defined in regulation 2, such distance being measured at right angles to the centreline at the level of the deepest subdivision draught, the operation of the watertight doors clear of the damaged portion of the ship is not impaired.

6 All power-operated slidi ng watertight doors shall be provided with means of indication which will show at all remote operating positions whet her the doors are open or closed. Remote operating pos itions shall only be at the navigation bridge as required by paragraph 7.1.5 and at the location where hand operation above the bulkhead deck is required by paragraph 7.1.4.

7.1 Each power-operated sliding watertight door:

.1 shall have a vertical or horizontal motion;

.2 shall, subject to pa ragraph 10, be normally limited to a maximum clear opening width of 1.2 m. The Administration may permit larger doors only

.1 shall have a vertical or horizontal motion;

.2 shall, subject to pa ragraph 10, be normally limited to a maximum clear opening width of 1.2 m. The Administration may permit larger doors only to the extent considered necessary fo r the effective operation of the ship provided that other safety measures, including the following, are taken into consideration:

.1 special consideration shall be given to the strength of the door and its closing appliances in order to prevent leakages; and

.2 the door shall be located inboard the damage zone B/5;

.3 shall be fitted with the necessary equipment to open and close the door using electric power, hydraulic power, or any other form of power that is acceptable to the Administration;

.4 shall be provided with an individua l hand-operated mechanism. It shall be possible to open and close the door by hand at the door itself from either side, and in addition, close the door from an accessible position above the

possible to open and close the door by hand at the door itself from either side, and in addition, close the door from an accessible position above the bulkhead deck with an all round crank motion or some other movement providing the same degree of safety acceptable to the Administration. Direction of rotation or other movement is to be clearly indicated at all operating positions. The time necessary for the complete closure of the door, when operating by hand gear, shal l not exceed 90 s with the ship in the upright position;

.5 shall be provided with controls for opening and closing the door by power from both sides of the door and also for closing the door by power from the central operating console at the navigation bridge;

.6 shall be provided with an audible alar m, distinct from any other alarm in the area, which will sound whenever the door is closed remotely by power and which shall sound for at least 5 s but no more than 10 s before the door begins

area, which will sound whenever the door is closed remotely by power and which shall sound for at least 5 s but no more than 10 s before the door begins to move and shall continue sounding until the door is completely closed. In the case of remote hand operation it is sufficient for the audible alarm to

sound only when the door is moving. Additionally, in passenger areas and areas of high ambient noise the Administration may require the audible alarm to be supplemented by an intermittent visual signal at the door; and

.7 shall have an approximately unifo rm rate of closure under power. The closure time, from the time the door be gins to move to the time it reaches the completely closed position, shall in no case be less than 20 s or more than 40 s with the ship in the upright position.

the completely closed position, shall in no case be less than 20 s or more than 40 s with the ship in the upright position.

7.2 The electrical power requ ired for power-operated sliding watertight doors shall be supplied from the emergency switchboard either directly or by a dedicated distribution board situated above the bulkh ead deck. The associated control, indication and alarm circuits shall be supplied from the emer gency switchboard either directly or by a dedicated distribution board situated above the bulkhead deck and be capable of being automatically supplied by the transitional sour ce of emergency electrical power required by regulation 42.3.1.3 in the event of failure of either the main or emergency source of electrical power.

7.3 Power-operated sliding watert ight doors shall have either:

.1 a centralized hydraulic system with two independent power sources each consisting of a motor and pump capable of simultaneously closing all

.1 a centralized hydraulic system with two independent power sources each consisting of a motor and pump capable of simultaneously closing all doors. In addition, there shall be for the whole installation hydraulic accumulators of sufficient capacity to operate all the doors at least three times, i.e. closed-open-closed, agains t an adverse list of 15º. This operating cycle shall be capable of being carried out when the accumulator is at the pump cut-in pressure. The fluid used shall be chosen considering the temperatures liable to be enc ountered by the installation during its service. The power operating system shall be designed to minimize the possibility of having a single failure in the hydraulic piping adversely affect the operation of more than one door. The hydraulic system shall be provided with a low-level alarm for hydr aulic fluid reservoirs serving the power-operated system and a low gas pressure alarm or other effective

provided with a low-level alarm for hydr aulic fluid reservoirs serving the power-operated system and a low gas pressure alarm or other effective means of monitoring loss of stored energy in hydraulic accumulators. These alarms are to be audible and visual and shall be situated on the central operating console at the navigation bridge; or

.2 an independent hydraulic system for each door with each power source consisting of a motor and pump capab le of opening and closing the door. In addition, there shall be a hydraulic accumulator of sufficient capacity to operate the door at least three times, i.e. closed-open-closed, against an adverse list of 15 o. This operating cycle shall be capable of being carried out when the accumulator is at the pump cut-in pressure. The fluid used shall be chosen considering the temperatures liable to be encountered by the installation during its service. A low gas pressure group alarm or other

shall be chosen considering the temperatures liable to be encountered by the installation during its service. A low gas pressure group alarm or other effective means of monitoring loss of stored energy in hydraulic accumulators shall be provided at the central operating console on the navigation bridge. Loss of stored energy indication at each local operating position shall also be provided; or

.3 an independent electrical system and motor for each door with each power source consisting of a motor capable of opening and closing the door. The power source shall be capable of being automatically supplied by the transitional source of emergency electrical power as required by

regulation 42.4.2 in the event of failu re of either the main or emergency

regulation 42.4.2 in the event of failu re of either the main or emergency source of electrical power and with suffi cient capacity to operate the door at least three times, i.e. closed-open-closed, against an adverse list of 15o.

For the systems specified in paragraphs 7.3.1, 7.3.2 and 7.3.3, provision should be made as follows: Power systems for power-operated watertight sliding doors shall be separate from any other power system. A single failure in the electric or hydraulic power-operated systems excluding the hydraulic actuator shall not prevent the hand operation of any door.

7.4 Control handles shall be provided at each side of the bulkhead at a minimum height of 1.6 m above the floor and shall be so arranged as to en able persons passing through the doorway to hold both handles in th e open position without being able to set

height of 1.6 m above the floor and shall be so arranged as to en able persons passing through the doorway to hold both handles in th e open position without being able to set the power closing mechanism in operation acci dentally. The direction of movement of the handles in opening and clos ing the door shall be in the direction of door movement and shall be clearly indicated.

7.5 As far as practicable, electrical equi pment and components for watertight doors shall be situated above the bulkhead deck and outside hazardous areas and spaces.

7.6 The enclosures of electrical components necessarily situated below the bulkhead deck shall provide suitable protection against the ingress of water.

7.7 Electric power, control, indication and alarm circuits shall be protected against fault in such a way that a failure in one door circuit will not cause a failure in any other door circuit. Short circuits or other faults in the alarm or indicator circuits of a door shall

fault in such a way that a failure in one door circuit will not cause a failure in any other door circuit. Short circuits or other faults in the alarm or indicator circuits of a door shall not result in a loss of power operation of that door. Arrange ments shall be such that leakage of water into the el ectrical equipment located below the bulkhead deck will not cause the door to open.

7.8 A single electrical failure in the power operating or control system of a power-operated sliding watertight door shal l not result in a closed door opening. Availability of the power supply should be continuously monitored at a point in the electrical circuit as near as practicable to each of the motors required by paragraph 7.3. Loss of any such power supply should activate an audible and visual alarm at the central operating console at the navigation bridge.

8.1 The central operating consol e at the navigation bridge shall have a master mode

operating console at the navigation bridge.

8.1 The central operating consol e at the navigation bridge shall have a master mode switch with two modes of cont rol: a local control mode which shall allow any door to be locally opened and locally closed after use without automatic closure, and a doors closed mode which shall automatically close any door that is open. The doors closed mode shall automatically close any door that is open. The doors closed mode shall permit doors to be opened locally and shall automatically re-close the doors upon release of the local control mechanism. The mas ter mode switch shall normally be in the local control mode. The doors closed mode shall only be used in an emergency or for testing purposes. Special consideration shall be given to the reliability of the master mode switch.

8.2 The central operating console at the na vigation bridge shall be provided with a

for testing purposes. Special consideration shall be given to the reliability of the master mode switch.

8.2 The central operating console at the na vigation bridge shall be provided with a diagram showing the location of each door, with visual indicators to show whether each door is open or closed. A red light shall i ndicate a door is fully open and a green light shall indicate a door is fully cl osed. When the door is closed remotely the red light shall indicate the intermediate position by flashing. The indicating circuit shall be independent of the control circuit for each door.

8.3 It shall not be possible to remo tely open any door from the central operating console.

9.1 If the Administration is satisfied that such doors are e ssential, watertight doors of

8.3 It shall not be possible to remo tely open any door from the central operating console.

9.1 If the Administration is satisfied that such doors are e ssential, watertight doors of satisfactory construction may be fitted in watertight bulkheads dividing cargo between deck spaces. Such doors may be hinged, rolling or sliding doors but shall not be remotely controlled. They shall be fitted at the highest level and as far from the shell plating as practicable, but in no case shall the outboard vertical edges be situated at a distance from the shell plating which is less than one fifth of the breadth of the ship, as defined in regulation 2, such distance being measured at ri ght angles to the centreline at the level of the deepest subdivision draught.

9.2 Should any such doors be accessible during the voyage, they shall be fitted with a device which prevents unauthorized opening. When it is proposed to fit such doors, the

9.2 Should any such doors be accessible during the voyage, they shall be fitted with a device which prevents unauthorized opening. When it is proposed to fit such doors, the number and arrangements shall receive the special consideration of the Administration.

10 Portable plates on bulkheads shall not be permitted except in machinery spaces. The Administration may permit not more than one power-operated sliding watertight door in each watertight bulkhead larger than those specified in paragraph 7.1.2 to be substituted for these portable plates, provide d these doors are intended to remain closed during navigation except in case of urgent necessity at the discretion of the master. These doors need not meet the requirements of paragraph 7.1.4 regarding complete closure by hand-operated gear in 90 s.

11.1 Where trunkways or tunnels for access from crew accommodation to the stokehold, for piping, or for any other purpos e are carried through watertight bulkheads,

hand-operated gear in 90 s.

11.1 Where trunkways or tunnels for access from crew accommodation to the stokehold, for piping, or for any other purpos e are carried through watertight bulkheads, they shall be watertight and in accordance with the requirements of regulation 16-1. The access to at least one end of each such tunnel or trunkway, if used as a passage at sea, shall be through a trunk extending watertight to a height sufficient to permit access above the bulkhead deck. The access to the other end of the trunkway or tunnel may be through a watertight door of the type required by its location in the ship. Such trunkways or tunnels shall not extend through the first subdivision bulkhead abaft the collision bulkhead.

11.2 Where it is proposed to fit tunnels pi ercing watertight bu lkheads, these shall receive the special consideration of the Administration.

11.3 Where trunkways in connection with refr igerated cargo and ventilation or forced

receive the special consideration of the Administration.

11.3 Where trunkways in connection with refr igerated cargo and ventilation or forced draught trunks are carried th rough more than one watert ight bulkhead, the means of closure at such openings shall be operated by power and be cap able of being closed from a central position situated above the bulkhead deck.

Regulation 13-1
Openings in watertight bulkheads and internal decks in cargo ships

1 The number of openings in watertight s ubdivisions is to be kept to a minimum compatible with the design and proper workin g of the ship. Where penetrations of watertight bulkheads and internal decks are necessary for access, piping, ventilation, electrical cables, etc., arrangements are to be made to maintain the watertight integrity. The Administration may permit relaxation in the watertightness of openings above the freeboard deck, provided that it is demonstrated that any progressive flooding can be

The Administration may permit relaxation in the watertightness of openings above the freeboard deck, provided that it is demonstrated that any progressive flooding can be easily controlled and that the safety of the ship is not impaired.

2 Doors provided to ensure the watertight integrity of internal openings which are used while at sea are to be sliding watertight doors capable of being remotely closed from the bridge and are also to be operable locally from each side of the bulkhead. Indicators are to be provided at the control position showing whether the doors are open or closed, and an audible alarm is to be provided at the door closure. The power, control and indicators are to be operable in the event of ma in power failure. Particular attention is to

and an audible alarm is to be provided at the door closure. The power, control and indicators are to be operable in the event of ma in power failure. Particular attention is to be paid to minimizing the eff ect of control system failure. Each power-operated sliding watertight door shall be provided with an in dividual hand-operated mechanism. It shall be possible to open and close the door by hand at the door itself from both sides.

3 Access doors and access hatch covers normally closed at sea, intended to ensure the watertight integrity of internal openings, shall be provided with means of indication locally and on the bridge showing whether these doors or hatch covers are open or closed. A notice is to be affixed to each such door or hatch cover to the effect that it is not to be left open.

4 Watertight doors or ramps of satisfactor y construction may be fitted to internally

A notice is to be affixed to each such door or hatch cover to the effect that it is not to be left open.

4 Watertight doors or ramps of satisfactor y construction may be fitted to internally subdivide large cargo spaces, provided that the Administration is satisfied that such doors or ramps are essential. These doors or ramps may be hinged, rolling or sliding doors or ramps, but shall not be remotely controll ed. Should any of the doors or ramps be accessible during the voyage, they shall be fitted with a device which prevents unauthorized opening.

5 Other closing appliances which are kept pe rmanently closed at sea to ensure the watertight integrity of internal openings sha ll be provided with a notice which is to be affixed to each such closing appliance to the eff ect that it is to be kept closed. Manholes fitted with closely bolted covers need not be so marked.

Regulation 14
Passenger ships carrying goods vehicles and accompanying personnel

fitted with closely bolted covers need not be so marked.

Regulation 14
Passenger ships carrying goods vehicles and accompanying personnel

1 This regulation applies to passenger ships designed or adapted for the carriage of goods vehicles and accompanying personnel.

2 If in such a ship the total numbe r of passengers which include personnel accompanying vehicles does not exceed 12 + Ad/25, where Ad = total deck area (square metres) of spaces available for the stowage of goods vehicles and where the clear height at the stowage position and at the entrance to such spaces is not less than 4 m, the provisions of regulations 13.9.1 and 13.9.2 in respect of watertight doors apply except that the doors may be fitted at any level in watertight bulkheads dividing cargo spaces. Additionally, indicators are required on the navigation bridge to show automatically when each door is closed and all door fastenings are secured.

Additionally, indicators are required on the navigation bridge to show automatically when each door is closed and all door fastenings are secured.

3 The ship may not be certified for a higher number of passengers than assumed in paragraph 2, if a watertight door has been fitted in accordance with this regulation.

Regulation 15
Openings in the shell plating below the bulkhead deck of passenger ships and the freeboard deck of cargo ships

1 The number of openings in the shell plating shall be reduced to the minimum compatible with the design and proper working of the ship.

RESOLUTION MSC.216(82)
(adopted on 8 December 2006)
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2 The arrangement and efficiency of the means for closing any opening in the shell plating shall be consistent with its intende d purpose and the position in which it is fitted and generally to the satisfaction of the Administration.

plating shall be consistent with its intende d purpose and the position in which it is fitted and generally to the satisfaction of the Administration.

3.1 Subject to the requirements of the In ternational Conventio n on Load Lines in force, no sidescuttle shall be fitted in such a position that its sill is below a line drawn parallel to the bulkhead deck at side and having its lowest point 2.5% of the breadth of the ship above the deepest subdivision draught, or 500 mm, whichever is the greater.

3.2 All sidescuttles the sills of which are below the bulkhead deck of passenger ships and the freeboard deck of cargo ships, as permitted by paragraph 3.1, shall be of such construction as will effectively prevent any person opening them without the consent of the master of the ship.

4 Efficient hinged inside deadlights so arranged that they can be easily and effectively closed and secured watertight, shall be fitted to all sidescuttles except that

the master of the ship.

4 Efficient hinged inside deadlights so arranged that they can be easily and effectively closed and secured watertight, shall be fitted to all sidescuttles except that abaft one eighth of the ship's length from the forward perpendicular and above a line drawn parallel to the bulkhead deck at si de and having its lowest point at a height of 3.7 m plus 2.5% of the br eadth of the ship above the de epest subdivision draught, the deadlights may be portable in passenger ac commodation other than that for steerage passengers, unless the deadlights are require d by the International Convention on Load Lines in force to be permanently attached in their proper positions. Such portable deadlights shall be stowed adjacent to the sidescuttles they serve.

5.1 No sidescuttles shall be fitted in any spaces which are appropriated exclusively to the carriage of cargo or coal.

deadlights shall be stowed adjacent to the sidescuttles they serve.

5.1 No sidescuttles shall be fitted in any spaces which are appropriated exclusively to the carriage of cargo or coal.

5.2 Sidescuttles may, however, be fitted in spaces appropriated alternatively to the carriage of cargo or passengers, but they shal l be of such construction as will effectively prevent any person opening them or their deadlights without the consent of the master.

6 Automatic ventilating sidescuttles shall not be fitted in the shell plating below the bulkhead deck of passenger ships and the fr eeboard deck of cargo ships without the special sanction of the Administration.

7 The number of scuppers, sanitary discharges and other similar openings in the shell plating shall be reduced to the mini mum either by making each discharge serve for as many as possible of the sanitary and other pipes, or in any other satisfactory manner.

shell plating shall be reduced to the mini mum either by making each discharge serve for as many as possible of the sanitary and other pipes, or in any other satisfactory manner.

8.1 All inlets and discharges in the shell plating shall be fitted with efficient and accessible arrangements for preventing the accidental admission of water into the ship.

8.2.1 Subject to the requirements of the International Conventio n on Load Lines in force, and except as provided in paragraph 8.3, each separate discharge led through the shell plating from spaces below the bulkhead deck of passenger ships and the freeboard deck of cargo ships shall be provided with either one automatic non-return valve fitted with a positive means of closing it from above the bulkhead deck or with two automatic non-return valves without positive means of closing, provided that the inboard valve is situated above the deepest subdivision dra ught and is always accessible for examination